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EXERCISE - 1 : REPRODUCTIVE PARTS OF A FLOWER

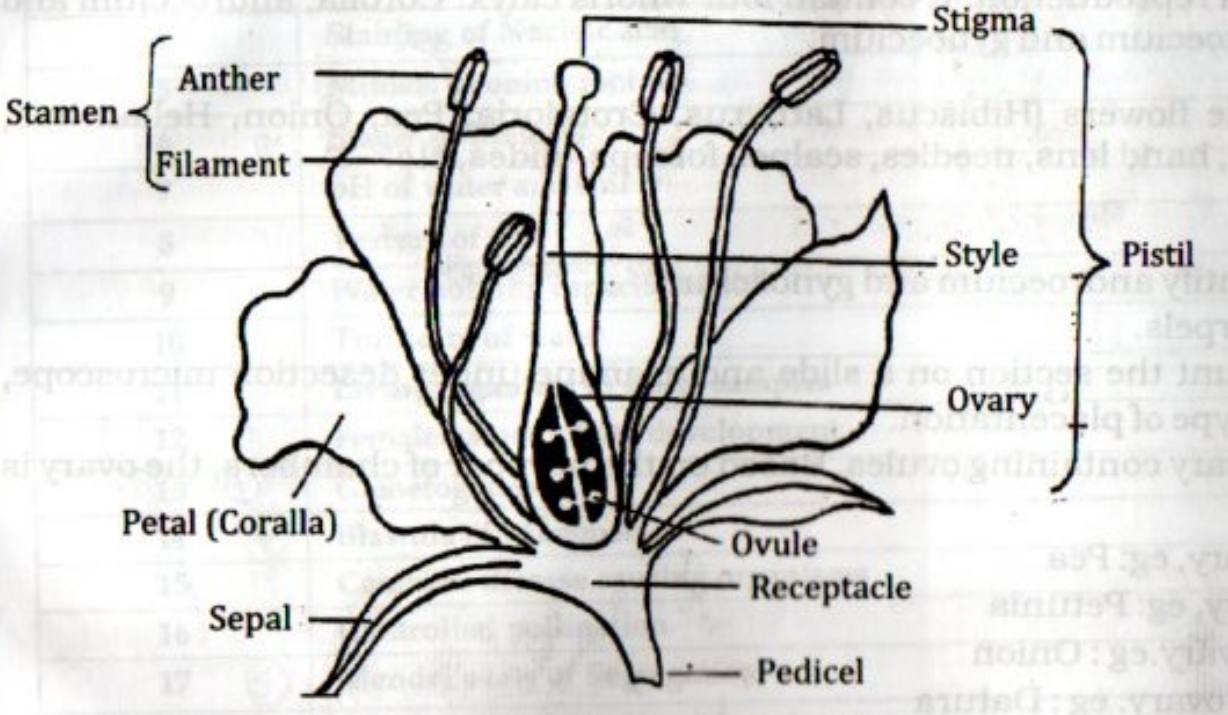
Aim : To study the reproductive parts of commonly available flowers

Principle : Flower is the site of sexual reproduction. It contain four whorls calyx. Corolla, androecium and gynoecium. Essential whorls are Androecium and gynoecium.

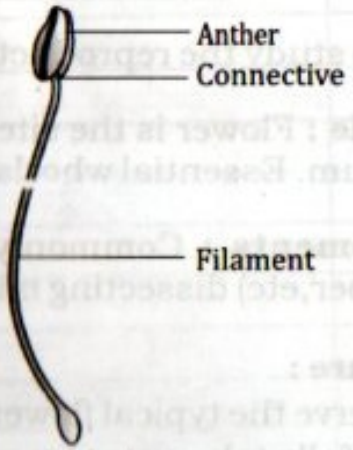
Requirements : Commonly available flowers (Hibiscus, Lathyrus, Crotaloria, Pea, Onion, Helianthus, Cucumber,etc) dissecting microscope, hand lens, needles, scalpel, forceps, slides, etc.

Procedure :

1) Observe the typical flower and identify androecium and gynoecium.
2) Carefully take out stamens and carpels.
3) Take a thin T/S of ovary and mount the section on a slide and examine under desection microscope, record the number of locules and type of placentation.
4) Locules are the chambers of the ovary containing ovules. Based on the number of chambers, the ovary is of following types.
   - **a) Unilocular :** one chambered ovary, eg: Pea
   - **b) Bilocular :** Two chambered ovary, eg: Petunia
   - **c) Trilocular :** Three chambered ovary.eg : Onion
   - **d) Tetralocular :** Four chambered ovary. eg : Datura
   - **e) Pentalocular :** Five chambered ovary, eg : Hibiscus
5) **Placentation :** It is the arrangement of ovules with in the ovary, it is of the following types.
   - **a) Marginal placentation -** Placenta bearing ovules develop near the margin on placenta. eg : pea.
   - **b) Axil placentation -** Ovules arises from the central axis of the ovary, eg : Hibiscus.
   - **c) Parietal placentation -** Ovules are arises from the inner wall of the ovary, eg - Mustard
   - **d) Free central placentation -** ovules are arises from the center of the
   - **e) Basal placentation -** Ovules are arises from the base of the unilocular ovary, eg - sunflower.

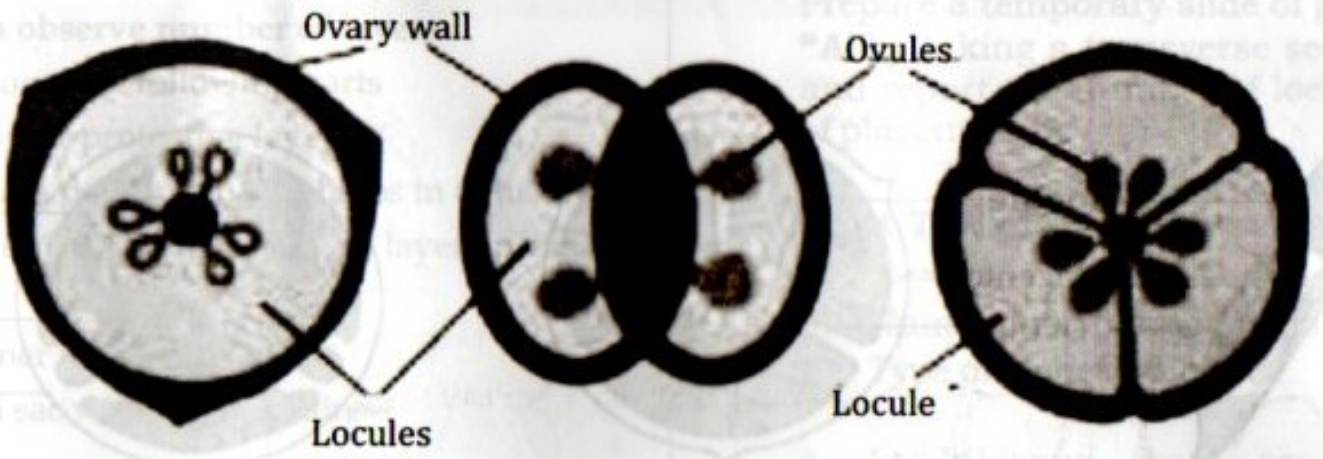


L.S OF FLOWER

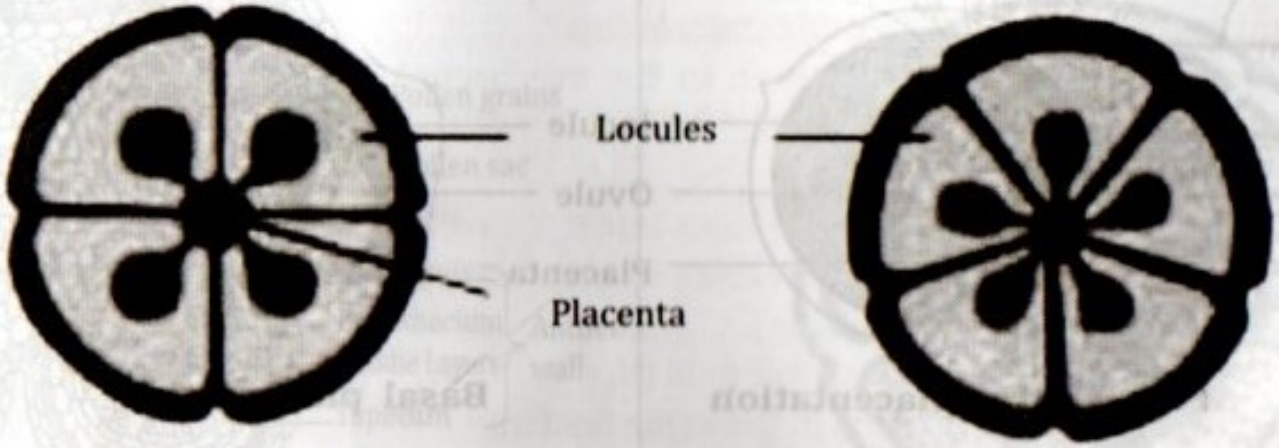


STAMEN

PISTIL

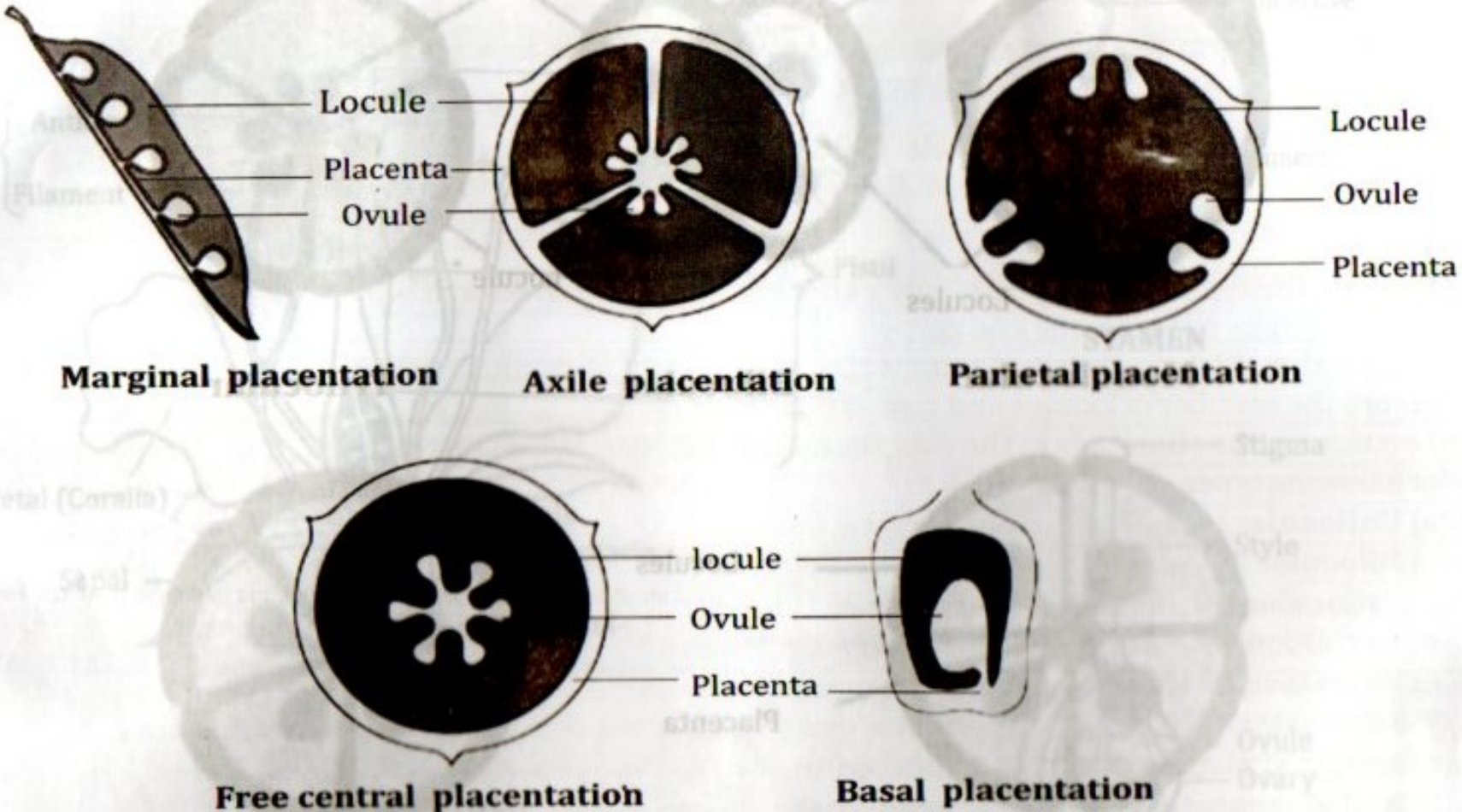


Monolocular **Bilocular** **Trilocular**



Tetralocular **Pentalocular**

NUMBER OF LOCULES

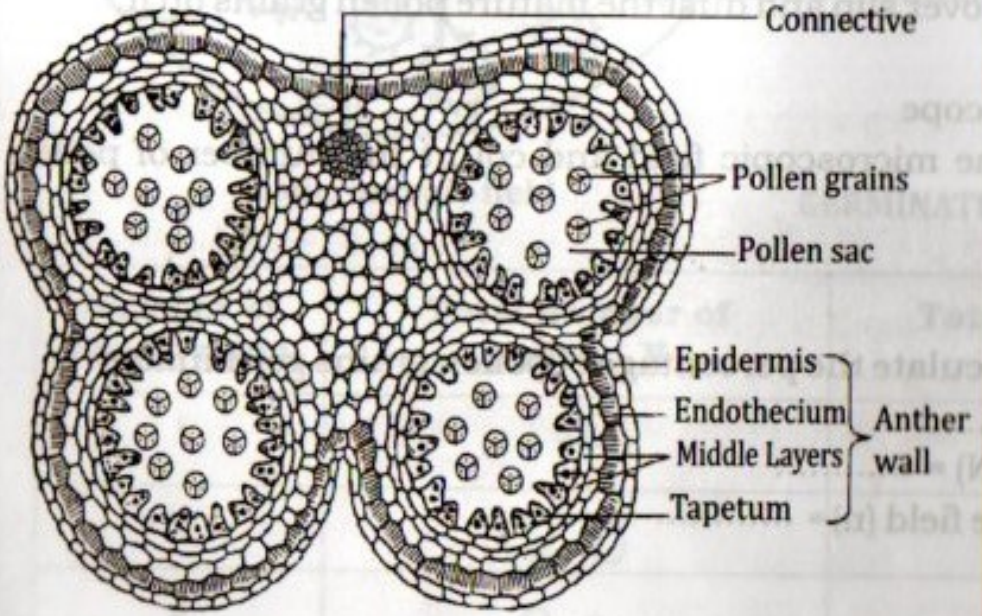


T.S. OF OVARY SHOWING PLACENTATION

T.S of Anther to observe number of layers

T.S. of anther shows the following parts

- **Epidermis :** outer protective layer
- **Endothecium** hypodermal layer helps in dehiscence
- **Middle layers :** 2-3 parenchymatous layers-protective in function
- **Tapetum :** Inner nourishing layer
- **Pollen Sac :** a sac with pollen grains



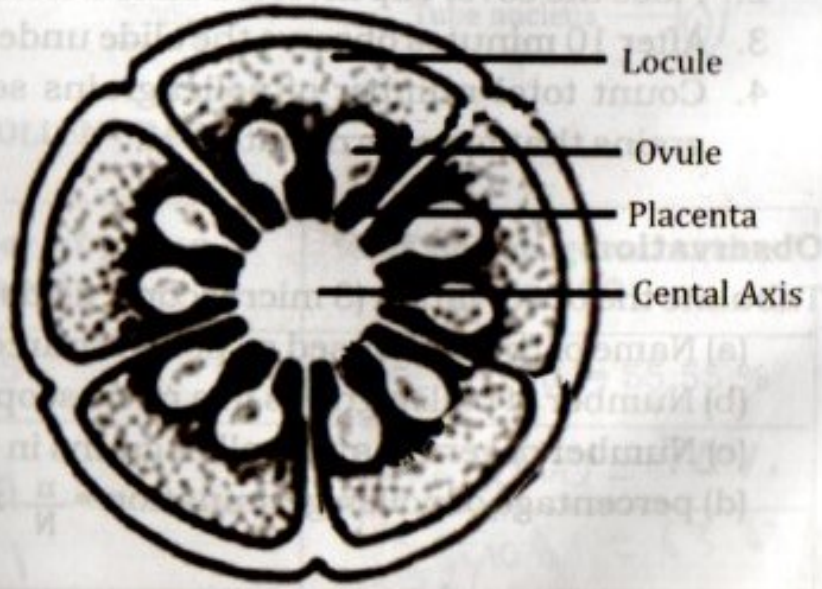
T.S. OF ANTHER

Prepare a temporary slide of given material **"A"**by taking a transverse section of ovary and report the number of locules and type of placentation.

Cut T.S. of ovary, mount it on a slide and observe

a. Number of locules in the ovary
b. Type of placentation

a. In hibiscus there are five locules **(Pentalocular)**
b. The type of placentation is **axile**



T.S. OF HIBISCUS OVARY

EXERCISE - 2 : POLLEN GERMINATION

Aim: To calculate the percentage of pollen germination

Principle: Artificially pollen grains are germinate in synthetic medium. During germination, the intine grows out as pollen tube through one of the germ pores in the exine.

Requirements: Mature pollen grains, 10% sucrose solution (10g sucrose dissolved in 100ml of H_20) Petridish, slides, cover slips, brush, needle and microscope etc.

Procedure :

1. Take a drop of 10% sucrose solution on a clean cover slip and dust the mature pollen grains on it.
2. Place the cover slip inverted on to a clean slide.
3. After 10 minutes observe the slide under microscope
4. Count total number of pollengrains seen in the microscopic field and count the number of pollen grains that have germinated.

Observation

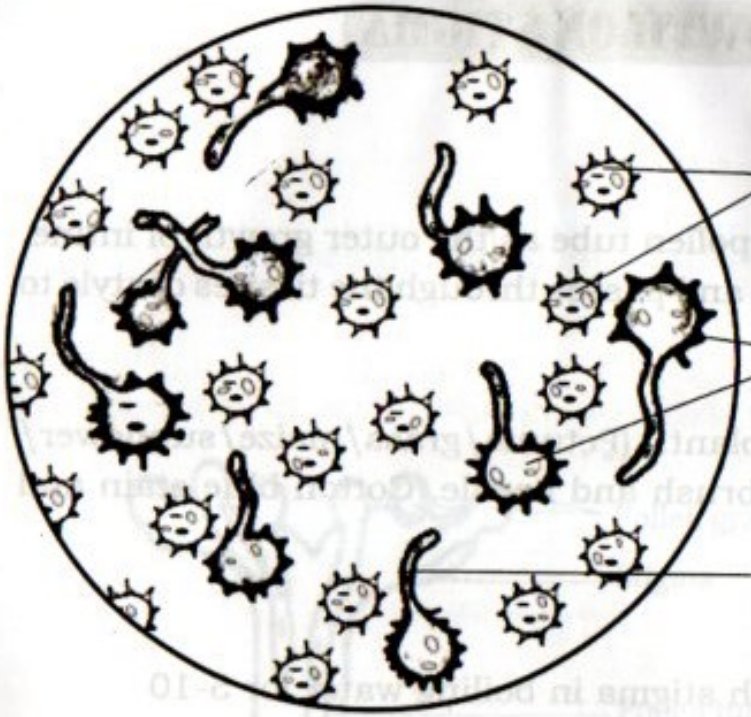
Tabulate the observation (3 microscopic fields) and calculate the percentage of pollen grains germinated.

(a) Name of the plant used as pollen source : .....

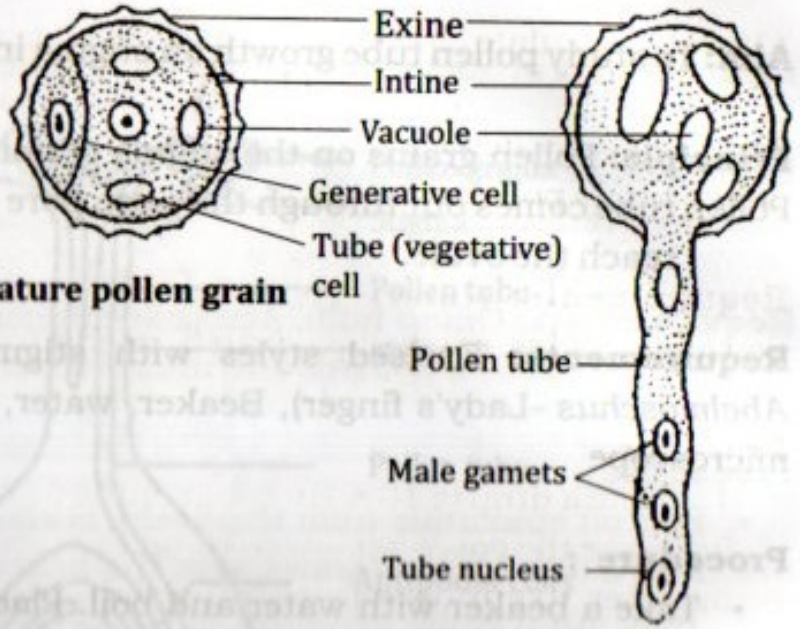
(b) Number of pollen grains in a microscopic field (N) = ...

(c) Number of germinated pollen grains in the same field (n)= .....

(d) percentage of pollen germination = $\frac{n}{N}$ x 100



Microscopic field



Mature pollen grain

GERMINATION OF POLLEN GRAINS

Number of observation	Total number of pollen - N	Total number of pollen germinated - n	% of pollen germination $\frac{n}{N}$ x 100
1	45	25	$\frac{25}{45}$ x 100 = 55.55 %
2	20	14	$\frac{14}{20} \times 100 = 70\%$
3	38	31	$\frac{38}{31} \times 100 = 63.63\%$
Average	46	31	$\frac{46}{31} \times 100 = 67.39\%$

EXERCISE - 3 : POLLEN TUBE GROWTH ON STIGMA

Aim: To study pollen tube growth on stigma in a pistil.

Principle: Pollen grains on the stigma germinated and produce pollen tube as the outer growth of intine. Pollen tube comes out through the germ pore present in the exine and passes through the tissues of style to reach the ovule.

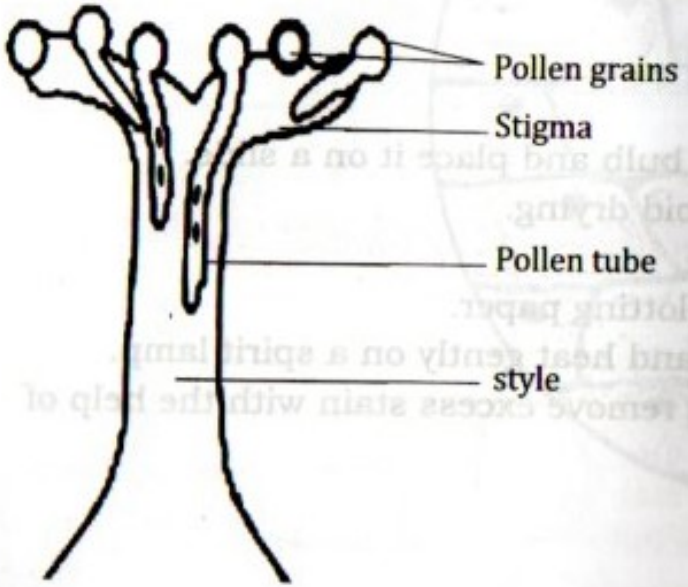
Requirements: Excised styles with stigma of a suitable plant. (Petunia/grass/maize/sunflower/ *Abelmoschus* -Lady's finger), Beaker, water, slides, coverslips, brush and needle, Cotton blue stain and microscope

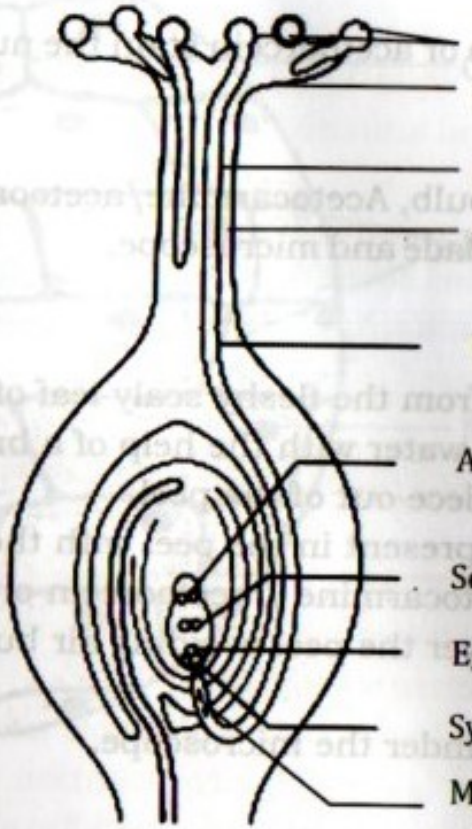
Procedure :

- Take a beaker with water and boil. Place the cut styles with stigma in boiling water for 5-10 minutes to soften the tissues.
- Take out a stigma and place on slide containing drop of water.
- Make an incision lengthwise through the style and stigma.
- Put a drop of cotton blue over it and allow to stand for 3-5 minutes
- Wash the excess stain away with water or blotting paper.
- Add a drop of glycerin / water and place a cover slip on it. Press the cover slip gently with a needle.
- If fail to observe pollen tubes mount another stigma.

Observation :

Blue - colored tubular moving through the tissues of style and stigmas are seen.





LONGITUDINAL SECTION OF A FLOWER SHOWING GROWTH OF POLLEN TUBE

EXERCISE - 4 : STAINING OF NUCLEIC ACID

Aim: Staining of nucleic acid by acetocarmine or acetoorcein.

Principle: Acetocarmine or acetoorcein stain the nucleic acid present in nucleic and present in nucleus to from deep red conjugate.

Requirements: Onion bulb, Acetocarmine/acetoorcein, blotting paper, slide, cover slips, brush, needles, scissors/ blade and microscope.

Procedure :

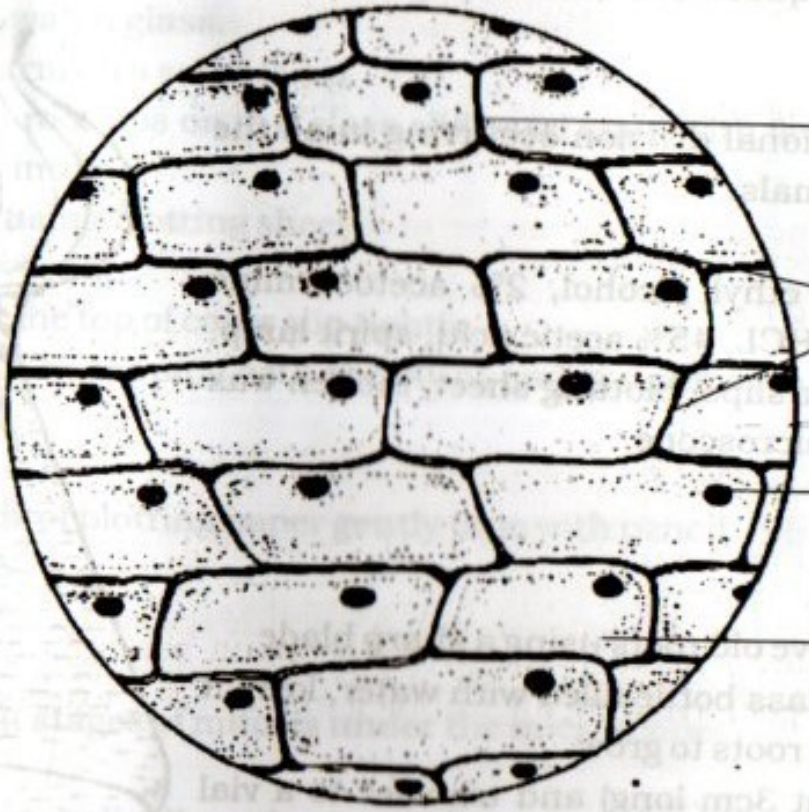
- Peel off epidermis from the fleshy scaly leaf of an onion bulb and place it on a slide.
- Add a few drops of water with the help of a brush to avoid drying.
- Take out a small piece out of the peel.
- Remove the water present in the peel with the help of blotting paper.
- Add 2 drops of acetocarmine / acetoorcein on the peel and heat gently on a spirit lamp.
- Place a coverslip over the peel avoiding air bubbles and remove excess stain with the help of blotting paper.
- Observe the cells under the microscope.

Observation :

Observe the shape of cell : _____

Number of nuclei in a cell : _____

Position of nuclei in the cell : _____



NUCLEAR STAINED PLANT CELLS

EXERCISE - 5 : MITOSIS IN ONION ROOT TIP

Aim: To prepare a temporary squash of onion root tip to observe different stages of mitosis.

Principle: Mitosis is an equational division occurring in all the somatic cells of plants and animals.

Requirements: Onion bulb, ethyl alcohol, 2% acetocarmine stain or acetoorcein stain, 1N HCL, 45% acetic acid, spirit lamp, watch glass, glass slides, cover slips, blotting sheet, molten wax or nail polish and compound microscope

Procedure :

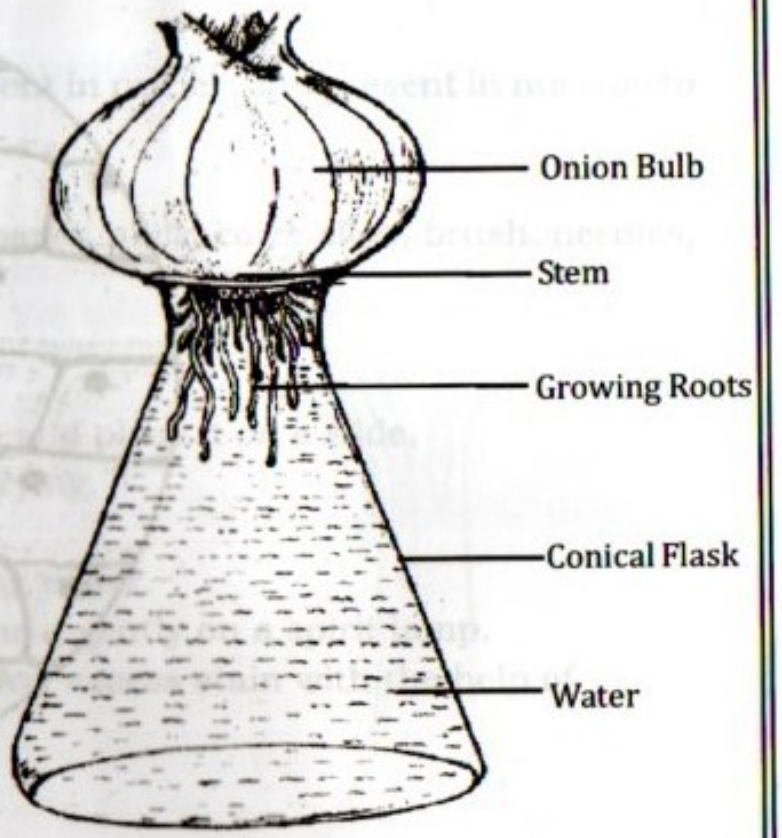
(1) Growing root tips:

(a) Take onion bulb and remove old roots using a sharp blade

(b) Place the onion bulb on glass bottle filled with water, keep it there about 5 days for new roots to grow

(c) Cut a few root tips (about 3cm long) and transfer to a vial containing freshly prepared fixative solution (1 part glacial acetic acid and 3 parts of ethy1 alcohol) for about 24 hour. This process is called fixation.

(d) Preserve the onion root tips in 70% alcohol



METHOD OF GROWING ONION ROOT

(2) Preparation of slide :

(a) Take onion root tips into a watch glass.
(b) Add 1 N HCL and gently warm on a spirit lamp.
(c) Transfer the treated onion root tips on to a glass slide and stain with few drops of stain acetocarmine / acetoorcein for about 10 minutes
(d) Remove excess of stain by using blotting sheet.
(e) Cover the material with cover slip.
(f) Place the blotting paper on the top of cover slip tightly.
(g) Gently squash the material by tapping with the blunt end of a pencil to spread the material into a thin film.
(h) Put cover slip on root tip
(i) Keep the slide between folds of blotting paper gently tape with pencil
(j) Observe under microscope

OBSERVATION:

Observe and record the different stages of mitosis under the microscope.

(1) Interphase:

- Resting phase nucleus metabolically active
- Preparatory under microscope

(2) Cell division :

- Two major events karyokinesis (division of nucleus) and cytokinessis (division of cytoplasm)
- Karyokinesis divided into four phases

(a) Prophase

- Chromatin becomes condensed thread like structures called chromosomes.
- Nuclear membrane and nucleolus starts disintegrating.

(b) Metaphase :

- The nuclear membrane disappears.
- Chromosomes are thick and are seen arranged at the equatorial plane of the cell.
- Each chromosome at this stage has two chromatids joined together at the centromere, which can be seen by changing the resolution of the microscope.
- Nucleolus is not observed during metaphase.

(c) Anaphase :

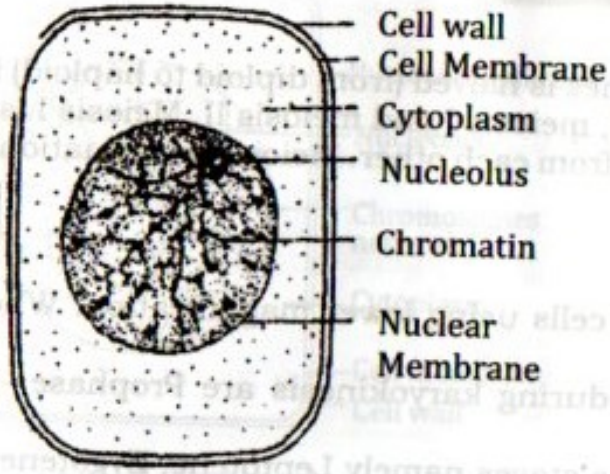
- This stage shows the separation of the chromatids of each chromosome.
- The chromatids separate due to the splitting of the centromere.
- Each chromatid now represents a separate chromosome as it has its own centromere.
- The chromosomes are found as if they have moved towards the two poles of the cell.
- The chromosomes at this stage may look like the shape of alphabets 'V', 'J' or 'I' depending upon the position of centromere in them.

(d) Telophase:

- Daughter chromosomes reach the opposite poles.
- Nuclear membrane and nucleoli reappear. Thus two daughter nuclei appear at opposite poles.

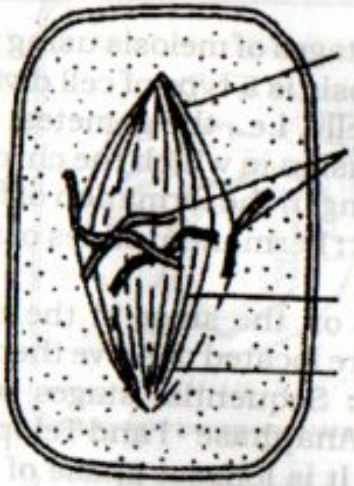
Cytokinesis :

After telophase a cleavage furrow appears around the equatorial plate which gradually deepens and divides the cytoplasm into two daughter cells.



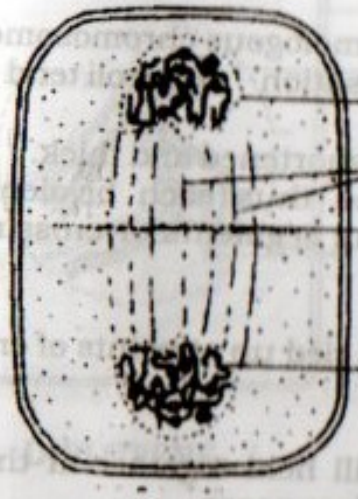
INTERPHASE

PROPHASE

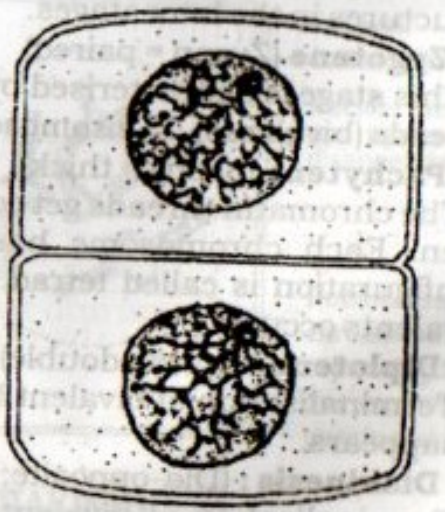


METAPHASE

ANAPHASE



TELOPHASE



DAUGHTER CELLS

EXERCISE - 6 : STAGES OF MEIOSIS

Aim: Study of stages of meiosis using permanent slides

Principle: Meiosis is a type of cell division in which the number of chromosomes is halved (from diploid to haploid) in the daughter cells, i.e., the gametes. The division is completed in two phases, meiosis I and meiosis II. Meiosis I is a reductional division in which the chromosomes of homologous pairs separate from each other. Meiosis II is equational division resulting in the formation of four daughter cells.

Requirements : Permanent slides of meiosis and compound microscope

Procedure :

Place the slide on the stage of the microscope and search for the dividing cells using lower magnification. When dividing cells are located observe them under higher magnification.

Meiosos - I : Sequential stages of Meiosis-I (Reductional divison) seen during karyokinesis are Prophase - I, Metaphase - I, Anaphase - I and Telophase - I

Prophase - I : It is longest phase of meiosis and it is further divided into five stages namely Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis.

(a) Leptotene (leptos = slender tene = band or thread)

The nuclear membrane and nucleolus are not distinctly observable Fine network of thin threads are seen uniformly distributed in the nucleus. These are chromatin threads, which may be observed as more prominent structures in the later stages.

(b) Zygotene (Zygon = paired)

This stage is characterised by the pairing of the homologous chromosomes, which can be seen as paired chromatin threads (bivalents) Chromosomes undergo condensation Nucleoli tend to disappear.

(c) Pachytene (pachy = thick)

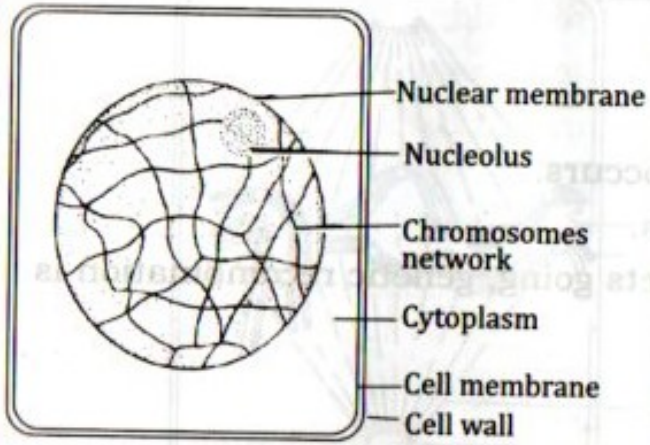
The chromatin threads get condensed and appear shortened and thick. Pairs of homologous chromosomes can be seen. Each chromosome has two chromatids and thus each bivalent consists of four chromatids. This configuration is called tetrad. Exchange of small segments or crossing over between non-sister chromatids of bivalents occurs

(d) Diplotene (Diplos - double):

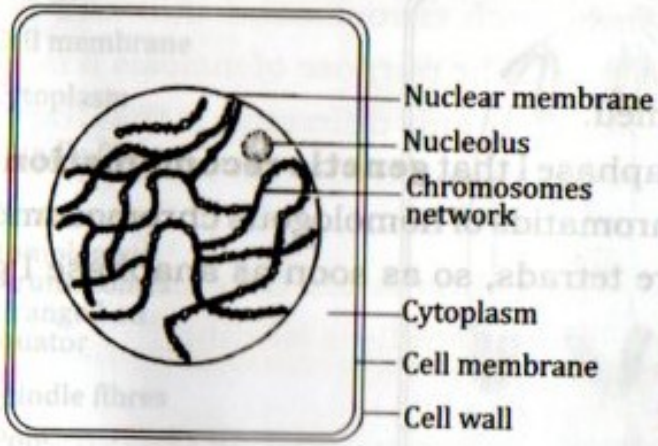
Terminalisation of bivalent begins Bivalents are tied up at points of crossing over called chiasmata. Nucleolus disappears.

(e) Diakinesis : (Dia-opposite; Kinesis - separation)

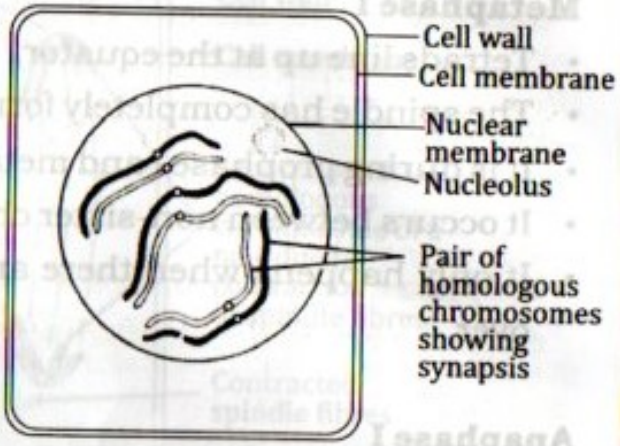
Terminalisation of bivalent continue and are still held together at the region of chiasmata Condensation of bivalent begins once again. Nuclear membrane disappears.



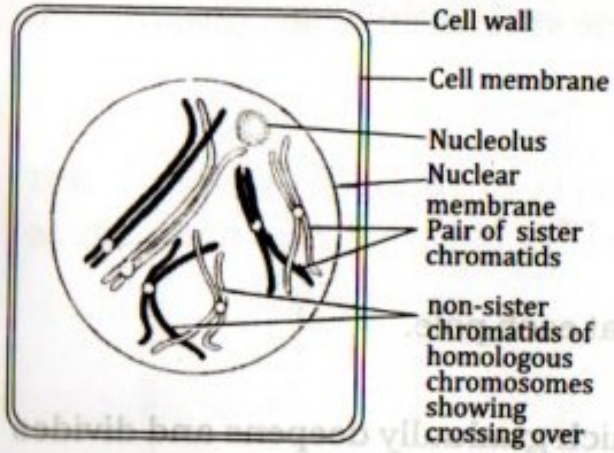
PRE LEPTOTENE



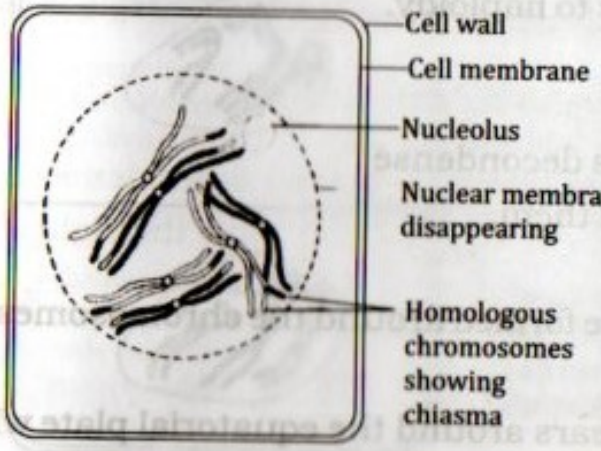
LEPTOTENE



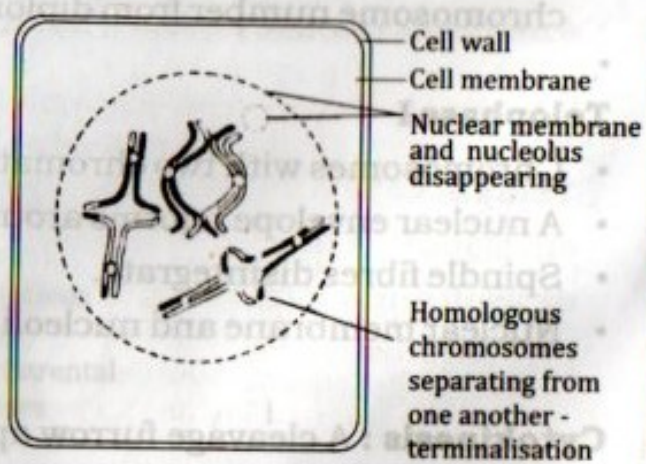
ZYGOTENE



PACHYTENE



DIPLOTENE



DIAKINESIS

MEIOSIS : PROPHASE I STAGES

Metaphase I

- Tetrads line up at the equator.
- The spindle has completely formed.
- It is during prophase I and metaphase I that **genetic recombination** occurs.
- It occurs between non-sister chromatids of homologous chromosomes.
- It only happens when there are tetrads, so as soon as anaphase I gets going, genetic recombination is over.

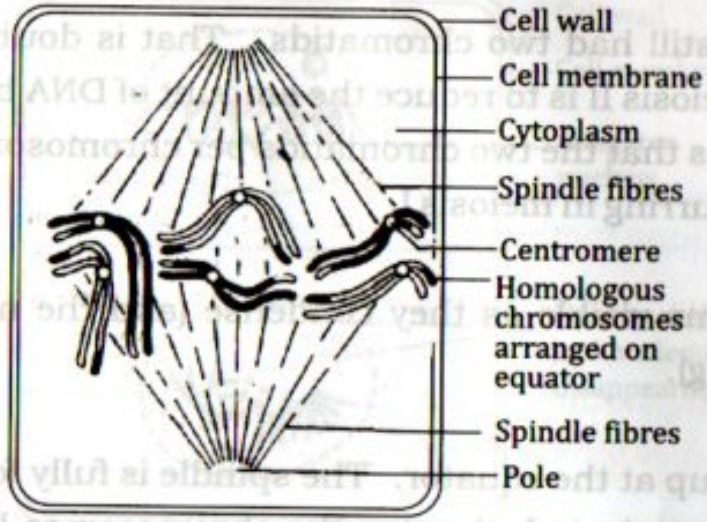
Anaphase I

- Tetrads pull apart and chromosomes with two chromatids move toward the poles.
- There is no splitting of centromere.
- Segregation of homologous chromosomes and their migration towards opposite poles cause reduction in chromosome number from diploidy to haploidy.
-

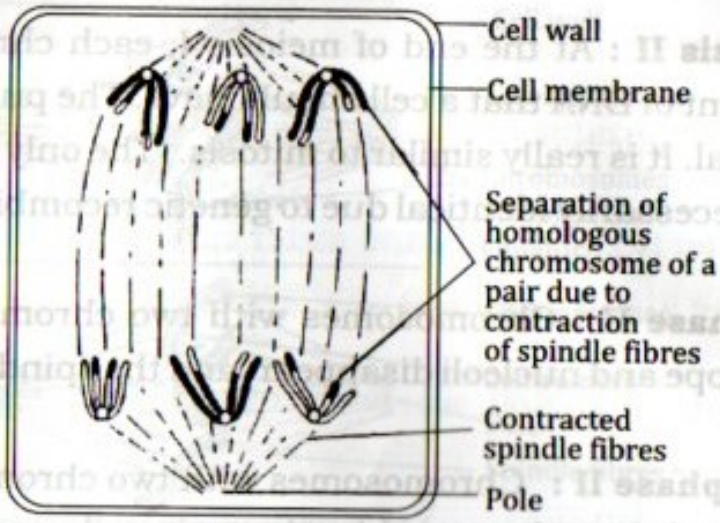
Telophase I

- Chromosomes with two chromatids decondense
- A nuclear envelope reforms around them.
- Spindle fibres disintegrate.
- Nuclear membrane and nucleoli are formed around the chromosomes at each pole.

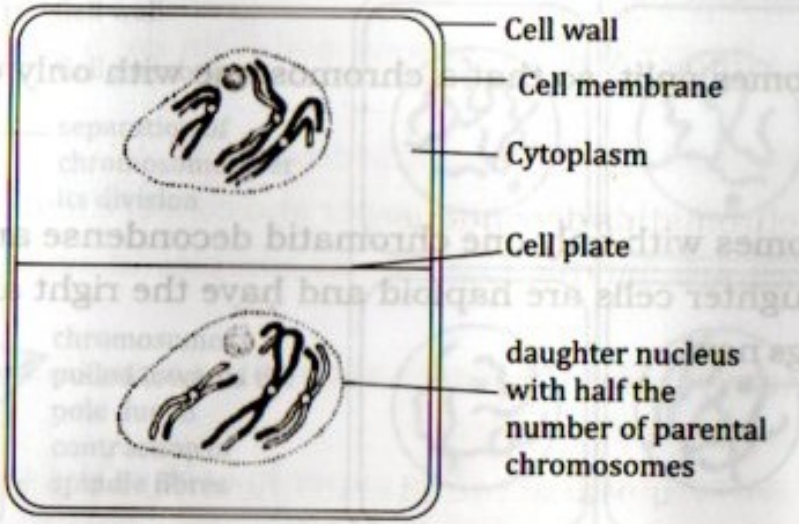
Cytokinesis : A cleavage furrow appears around the equatorial plate which gradually deepens and divides the cytoplasm into two daughter cells.



METAPHASE I



ANAPHASE I



TELOPHASE I

STAGES OF MEIOSIS - I

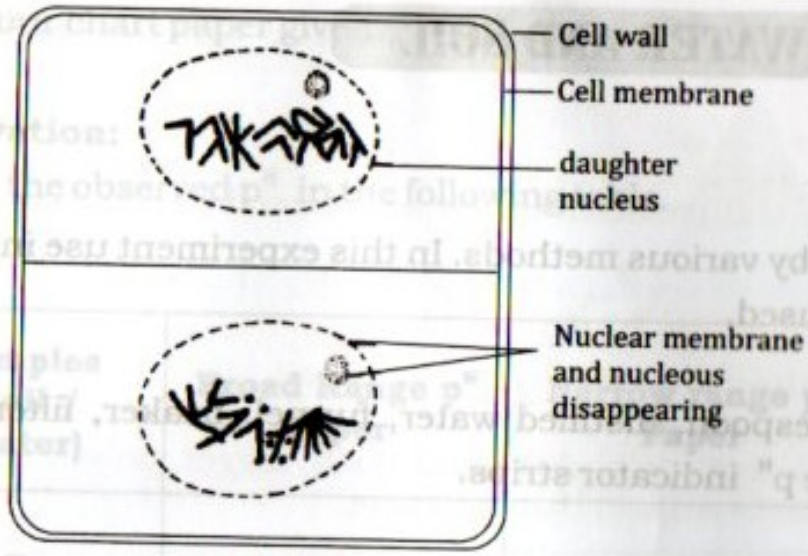
Meiosis II : At the end of meiosis I, each chromosome still had two chromatids. That is double the amount of DNA that a cell should have. The purpose of meiosis II is to reduce the amount of DNA back to normal. It is really similar to mitosis. The only difference is that the two chromatids per chromosome are not necessarily identical due to genetic recombination occurring in meiosis I.

Prophase II : Chromosomes with two chromatids become visible as they condense (and the nuclear envelope and nucleoli disappear, and the spindle is forming).

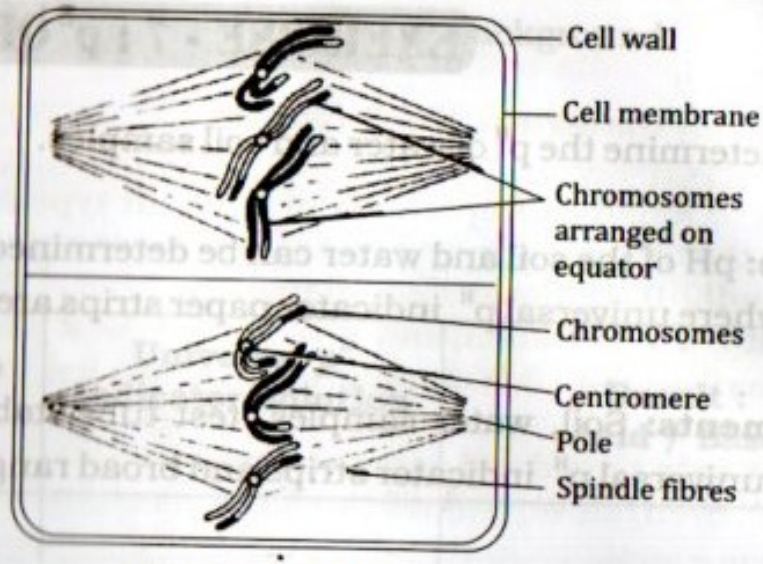
Metaphase II : Chromosomes with two chromatids line up at the equator. The spindle is fully formed. Although genetic recombination primarily occurs during meiosis I, the way the chromosomes line up during metaphase II can also help to make unique daughter cells.

Anaphase II : Chromosomes split, so that a chromosome with only one chromatid heads toward each pole.

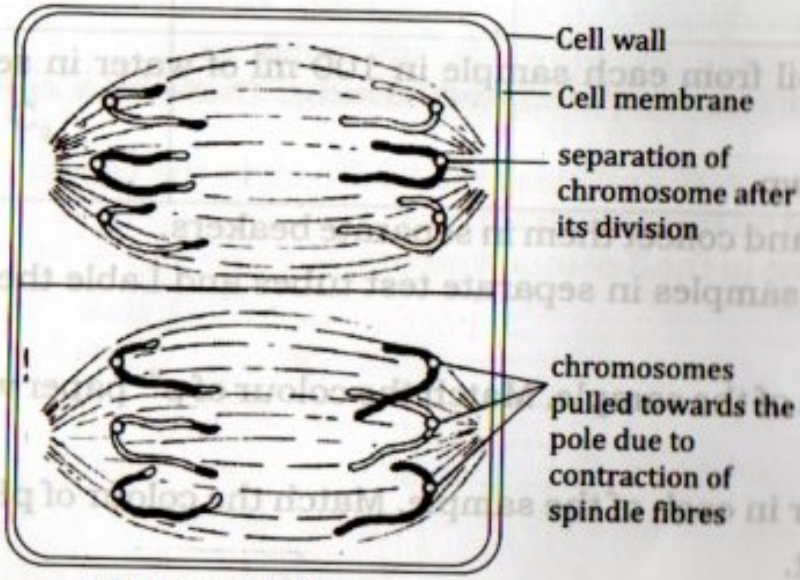
Telophase II : Chromosomes with only one chromatid decondense and get surrounded by new nuclear envelopes. The four daughter cells are haploid and have the right amount of DNA. They are ready to develop into sperm or eggs now.



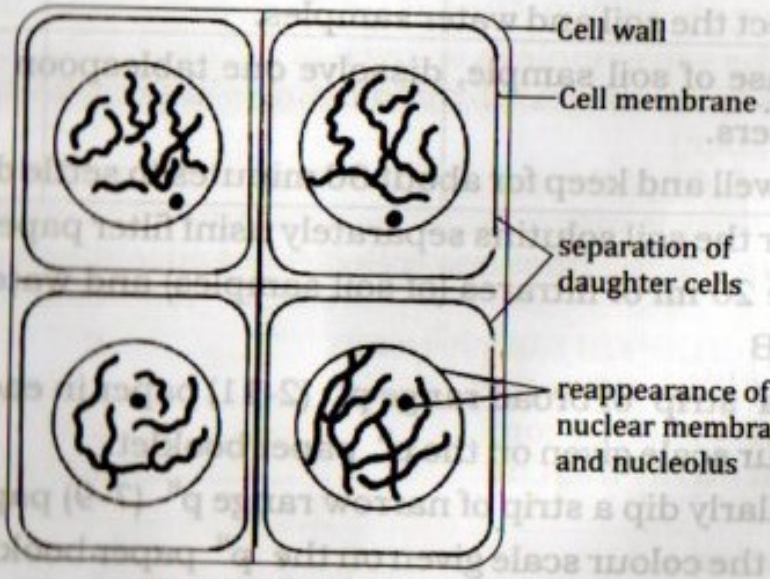
PROPHASE II



METAPHASE II



ANAPHASE II



TELOPHASE II :

STAGES OF MEIOSIS - II

EXERCISE - 7 : p^H OF WATER AND SOIL

Aim: To determine the p^H of water and soil samples.

Principle: pH of the soil and water can be determined by various methods. In this experiment use indicator method where universal p^H indicator paper strips are used.

Requirements: Soil, water samples, test tube, tablespoon, distilled water, funnel, beaker, filter paper dropper, universal p^H indicator strips and broad range p^H indicator strips.

Procedure :

1. Collect the soil and water samples.
2. In case of soil sample, dissolve one tablespoon soil from each sample in 100 ml of water in separate beakers.
   Stir well and keep for about 30 minutes to settle down,
3. Filter the soil solutins separately usinf filter paper and collect them in separate beakers.
4. Take 20 ml of filtrares (of soil samples) and water samples in separate test tubes and Lable them as A and B
5. Dip a strip of broad range p^H (2-11) paper in each of the sample. Match the colour of p^H paper with the colour scale given on the p^H paper booklet.
6. Similarly dip a strip of narrow range p^H (7-9) paper in each of the sample. Match the colour of pH paper with the colour scale given on the p^H paper booklet.
7. Add a few drops of universal pH indicator solution to each of the sample. Compare the colour with the

colour chart paper given on the universal p^{H} indicator bottle again st a white background.

Observation:

Record the observed p^{H} in the following table.

Samples (Soil / Water)	Broad Range p^{H} Paper	Narrow range p^{H} Paper	Universal p^{H} Indicator Solution	Result : Acid / Base
C_1				
C_2				

EXERCISE - 8 : TEXTURE OF SOIL

Aim : To study the texture of soil samples.

Principle : Relative proportions of different types of soil particles in the soil. Based on the sizw of soil particles, the texture is described as three types namely

Sand(2-0.05 mm average diameter),
Silt(0.05-0.002mm) and
Clay(less than 0.002mm).the soil which has all these fractions called loam soil.

Requirements : dried soil samples from two places (garden soil and roadside soil), digger, polythene bags, hand lens, oven/stove, balance, weights, mechanical sleve set and blotting sheet.

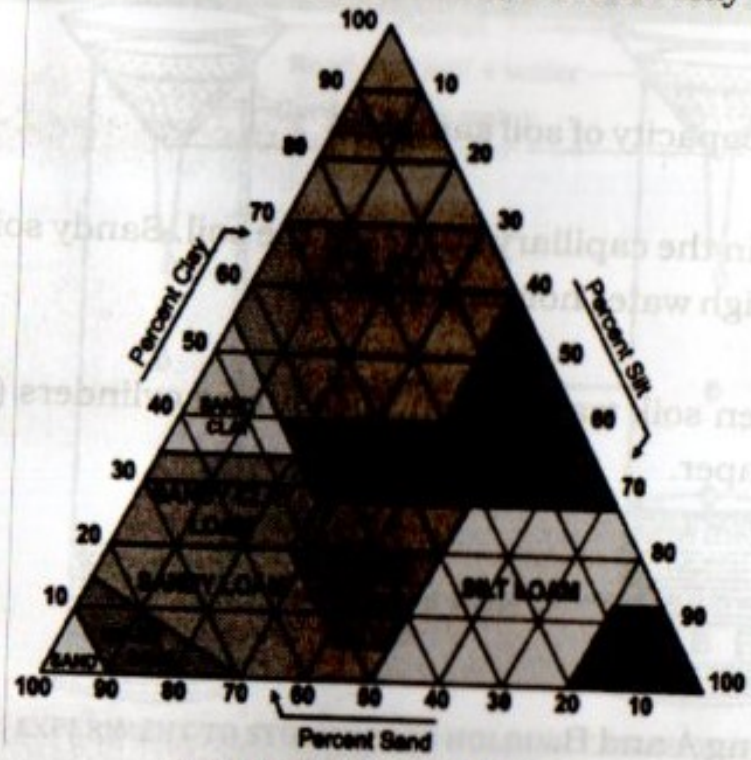
Procedure:

1. Collect about 300-500g of soil from garden and roadside in polythene bags. lable them as A and B.
2. Dry the soil sample in an oven or on stove or out in the sun to remove the soil moisture.
3. Take three sieves of different size (2mm, 0.5mm and 0.02mm) and arrange them in a collecting chamber
4. Take 200g pf the soil in the first sieve (sieve of 2mm mesh) and close the lid. To sieve the soil, shake the set of sieves manually for 5-10 minutes and collect the three soil fractions.
5. Weigh the soil fractions collected (sand, slit and clay) in the 3 compartments.

	A	B
• Weight of the soil sample taken	: _____g	_____g
• Weight of sand fraction	: _____g	_____g
• Weight of silt fraction	: _____g	_____g
• Weight of clay fraction	: _____g	_____g

Observation :
Calculate the percentage of sand, silt and clay fraction and tabulate.

Observation : Calculate the percentage of sand, silt and clay fraction and tabulate.



TEXTURAL TRIANGLE

Soil Sample	Percentage %			Texture class
	Sand	Silt	Clay	
A	21.33	11.93	66.73	clay
B				

EXERCISE - 9 : WATER HOLDING CAPACITY OF SOIL

Aim: To determine the water holding capacity of soil samples.

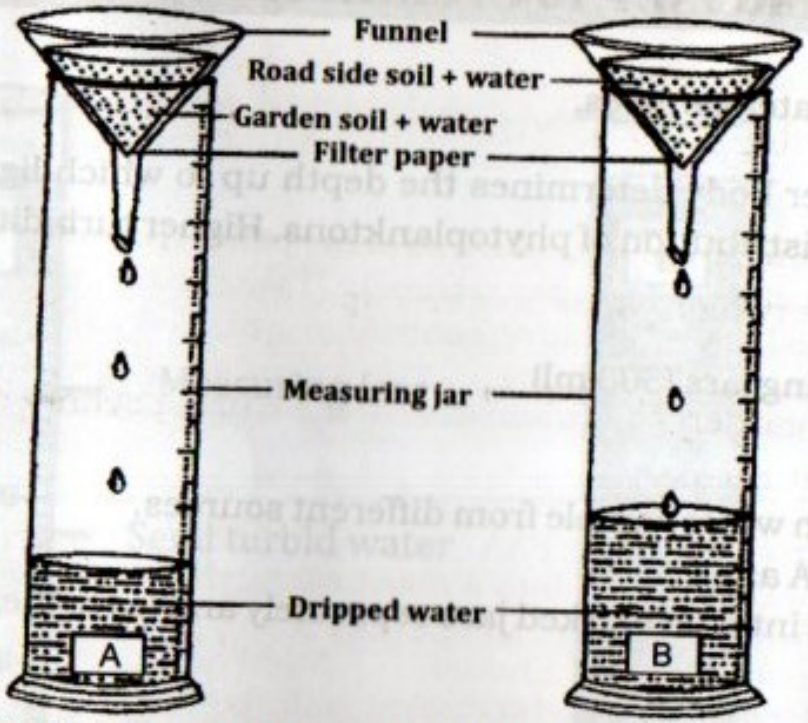
Principle: Amount of water retained in the capillary spaces of the soil. Sandy soil has very low water holding capacity, whereas clay soil has very high water holding capacity.

Requirements : Roadside soil, garden soil, water, oven, measuring cylinders (250ml), filter paper, muslin cloth, beaker, balance and blotting paper.

Procedure :

- Clean the soil samples.
- Dry the cleaned soil samples.
- Label the measuring cylinders using A and B.
- Place a funnel with filter paper on each of the measuring cylinders.
- Put 100 g of dried soil sample each into funnels placed on the measuring jars.
- Pour 100 ml of water over the soil samples in each funnel.
- Record the volume of filtered out water in the measuring cylinder when the dripping of water stops form the funnel.

Observation :



EXPERIMENT TO STUDY WATER HOLDING CAPACITY OF SOILS

Soil Sample	Wt. of soil (Z)g	Water added to soil (A) ml	Dripped water collected (B) ml	Water retained by soil (A-B) ml	Water holding capacity of soil A-B/Z ml/g
A. Road side soil	100	100	60	40	0.4 ml/g
B. Garden soil					

EXERCISE - 10 : TURBIDITY OF WATER

Aim : To study the turbidity of water samples.

Principle : Turbidity of the water body determines the depth up to which light can penetrate and thus it affects the photosynthesis and distribution of phytoplanktons. Higher turbidity affects the photosynthesis.

Requirements :

Two water samples, two measuring jars (500 ml)

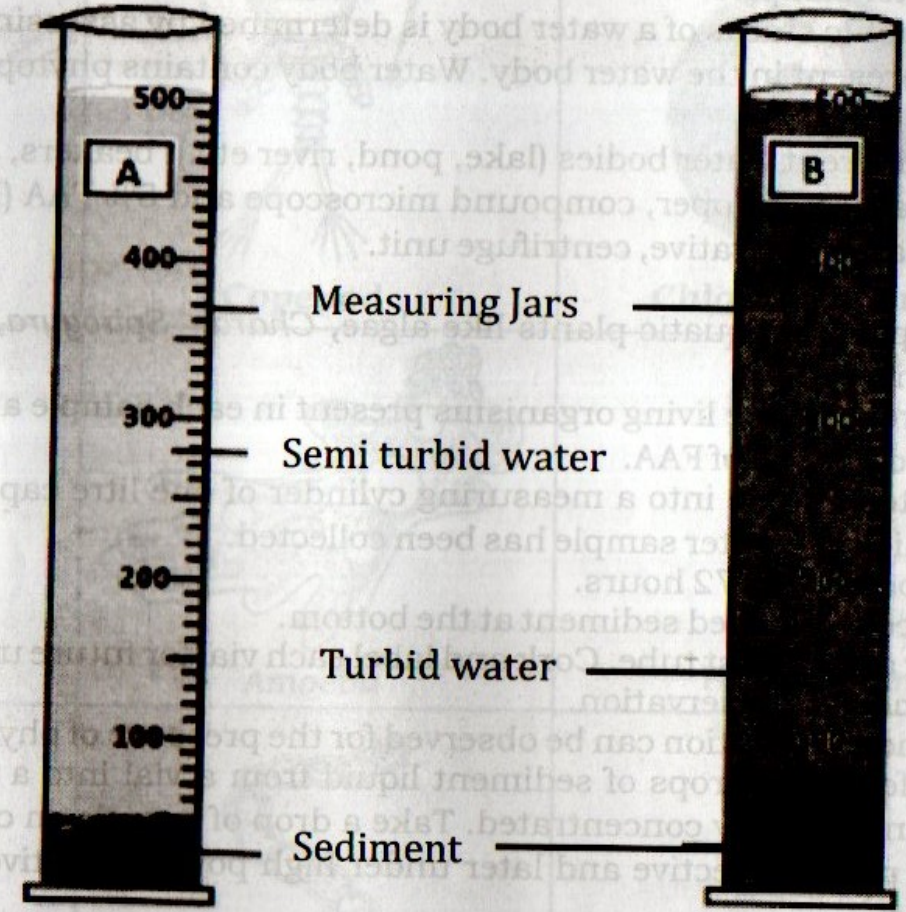
Procedure :

- Collect about two liters of each water sample from different sources.
- Mark the two measuring jars A and B
- Pour 500 ml of water samples into the marked jars separately and leave them undisturbed overnight.

Observations :

Observe the amount of sediment settled at the bottom of each jar and also note whether the water above the sediment is still turbid.

Water sample	Thickness of sediment (ml)	Clarity of water turbid/semiturbid/clear
A		
B		



TURBIDITY OF WATER SAMPLES

EXERCISE - 11 : LIVING ORGANIMS IN WATER SAMPLES

Aim: To analyze living organisms in water samples

Principle: The productivity and the trophic status of a water body is determined by assessing the number and type of organisms (micro as well as macro) present in the water body. Water body contains phytoplankton as the primary produces in aquatic ecosystems.

Requirements: Water samples from different water bodies (lake, pond, river etc.), beakers, a few vials or small test tubes, slides and cover slips, watch glasses, dropper, compound microscope and 5% FAA (Formalin Aceto Alcohol 5:5:90:Formalin: Acetic acid: Ethanol) as preservative, centrifuge unit.

Procedure :

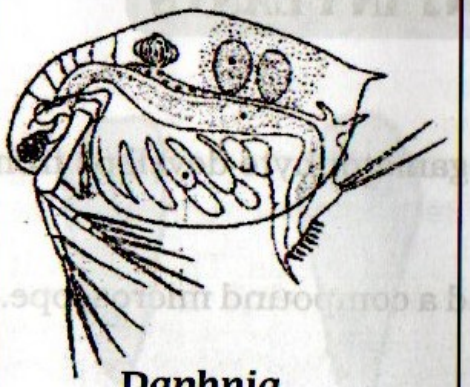
1. Collect about a liter of water sample and aquatic plants like algae, *Chara*, *Spirogyra*, *Zygnema* from nearby water body (pond, lake, reservoir, river).
2. Add about 5 ml of FAA to fix and preserve the living organisms present in each sample at the place of collection, 1% to 3% formaldehyde can be used instead of FAA.
3. In the laboratory, transfer the water sample into a measuring cylinder of one litre capacity. Label each water sample to indicate the site from which the water sample has been collected.
4. Leave the water samples undisturbed for 48-72 hours.
5. Decant off the clear water, leaving concentrated sediment at the bottom.
6. Transfer the sediment into a vial or a small test tube. Cork and label each vial for future use.
7. Centrifugation can be done for immediate observation.
8. The aquatic plants are squeezed and the solution can be observed for the presence of phytoplankton
9. With the help of a dropper, transfer a few drops of sediment liquid from a vial into a watch glass. Dilute the sediment with water if the sediment is highly concentrated. Take a drop of sample on clean grass slide put the cover slip first observe under low power objective and later under high power objective. The annexure can be used for identification of organisms.

Observation :

1. Record the different types of organisms present.
2. Count the number of organisms under each field of microscope.

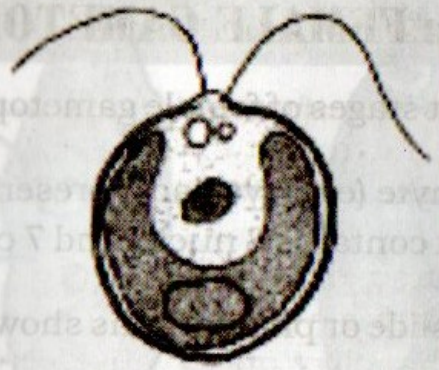
Discussion :

Prepare a list of organisms observed in each water sample and make an assessment of type and density of different organisms in each water sample.

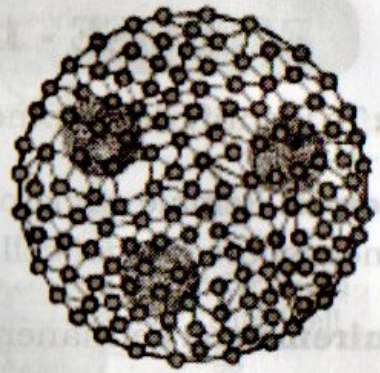


Daphnia

Copepod

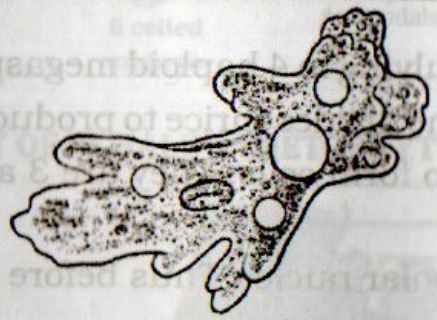


Chlamydomonas



Volvox

Euglena



Amoeba

Paramecium

Vorticella

Spirulina

Nostoc

Spirogyra

Rotifer

EXERCISE - 12 : FEMALE GAMETOPHYTE DEVELOPMENT IN PLANTS

Aim: To identify development stages of female gametophyte in ovary of a flower.

Principle: Female gametophyte (embryo sacs) present inside the ovule. Female gametophyte develops from the megaspore mother cell. It contains 8 nuclei and 7 cells

Requirements: Permanent slide or photographs showing female gametophyte and a compound microscope.

OBSERVATION

(1) Development of female gametophyte:

a. The megaspore mother cell undergoes meiosis resulying in 4 haploid megaspores

b. Out of which 3 disintergrates and the remaining one divides thrice to produce 8 haploid nuclei.

c. Of these 8 nuclei, 6 get surrounded by cell wall to form cells. They are 3 antipodals at chalazal end 2 synergids with an egg cell at the micropylar end.

d. The remaining 2 nuclei remain in the center as polar nuclei. Thus before fertilization the embroy sac has 8 nuclei and 7 cells.

2) V.S. of ovule (megasprangium)

a. Ovule develops inside the ovary.

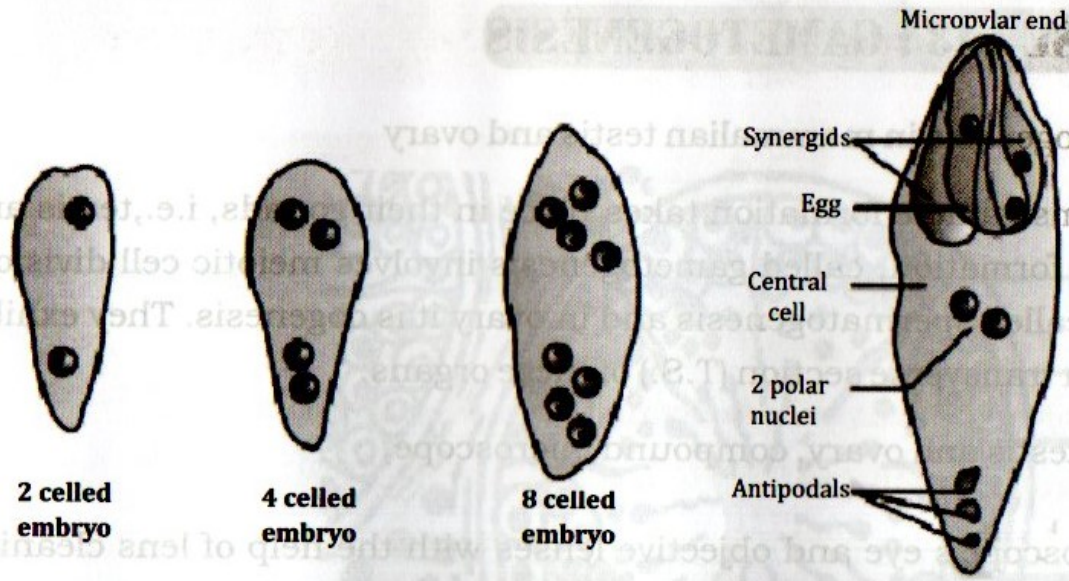
b. It is attached to the placenta by stalk funicle.

c. It consists of two integuments and nucellus.

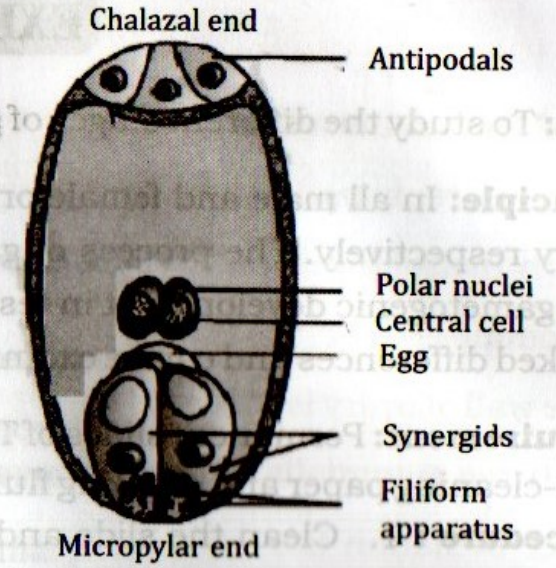
d. Nucellus cells contain food material.

e. Female gametophyte develops inside the nucellus.

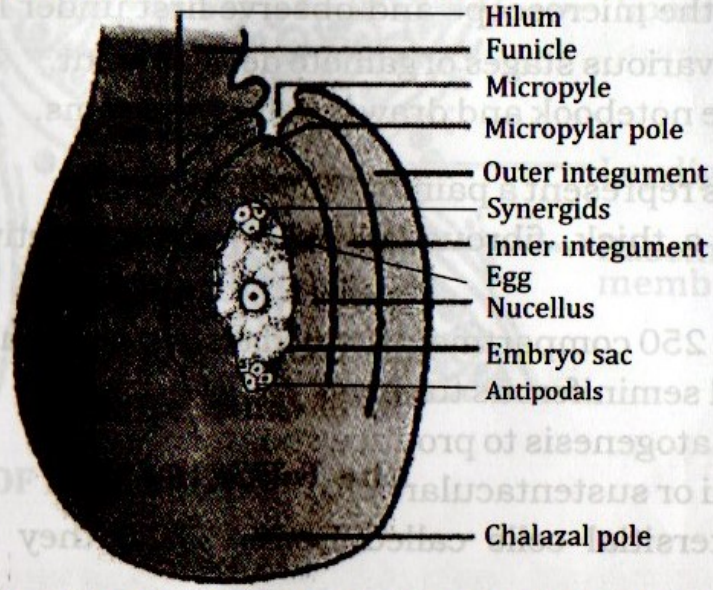
f. The embryo sac forms the megaspore mother cell.



STAGES OF DEVELOPMENT OF FEMALE GAMETOPHYTE IN ANGIOSPERMS



MATURE EMBRYO SAC



L.S. OF OVULE SHOWING FEMALE GAMETOPHYTE (EMBRYO SAC)

EXERCISE - 13 : GAMETOGENESIS

Aim: To study the different stages of gametogenesis in mammalian testis and ovary

Principle: In all male and female organisms gamete formation takes place in their gonads, i.e.,testis and ovary respectively. The process of gamete formation, called gametogenesis involves meiotic cell division. The gametogenic development in testis is called spermatogenesis and in ovary it is oogenesis. They exhibit marked differences and can be examined in transverse section (T.S.) of these organs.

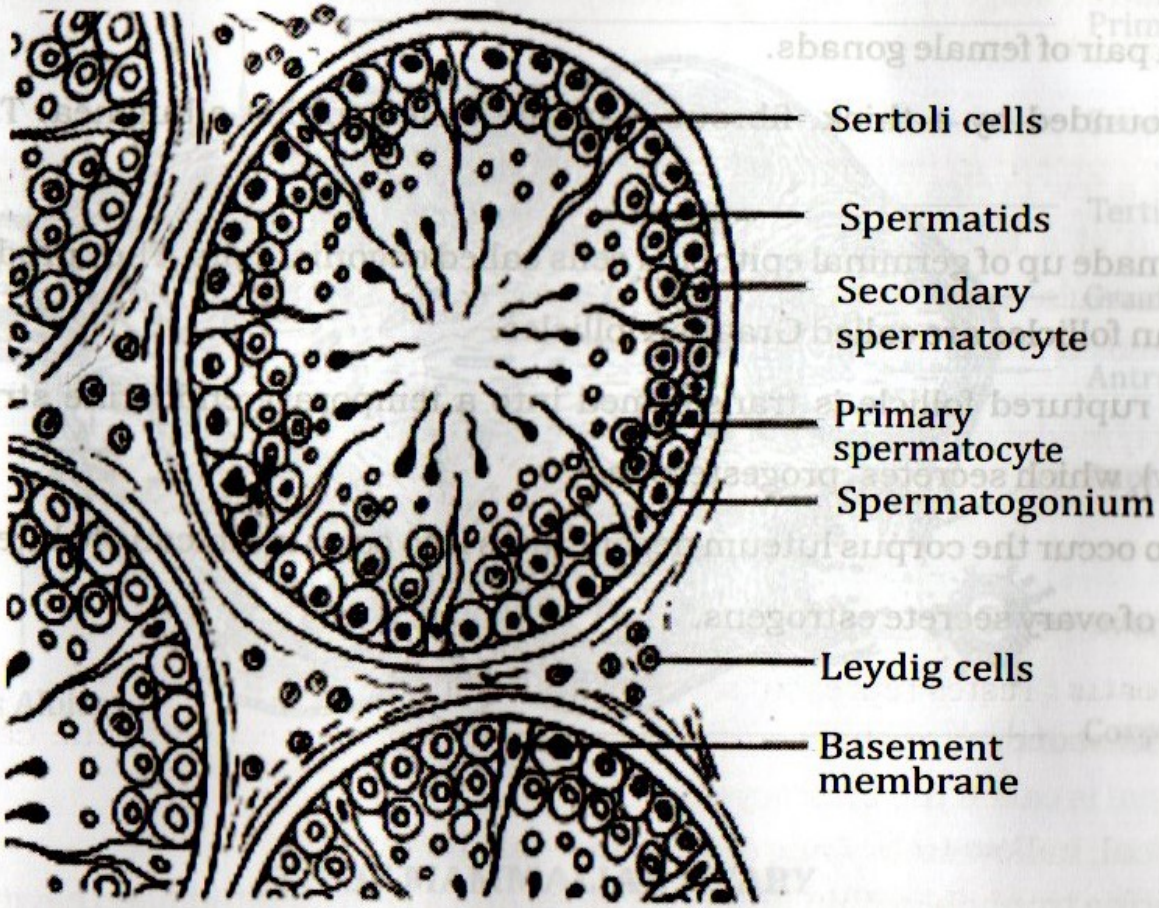
Requirement: Permanent slides of T.S. of testis and ovary, compound microscope, lens-cleaning paper and cleaning fluid

Procedure : 1. Clean the slide and microscope's eye and objective lenses with the help of lens cleaning paper using any cleaning fluid.

2. Place the slide on the stage of the microscope and observe first under lower magnification and then in higher magnification. Observe various stages of gamete development.
3. Record your observations in the notebook and draw labelled diagrams.

T.S of Mammalian Testis : Testes represent a pair of male gonads.

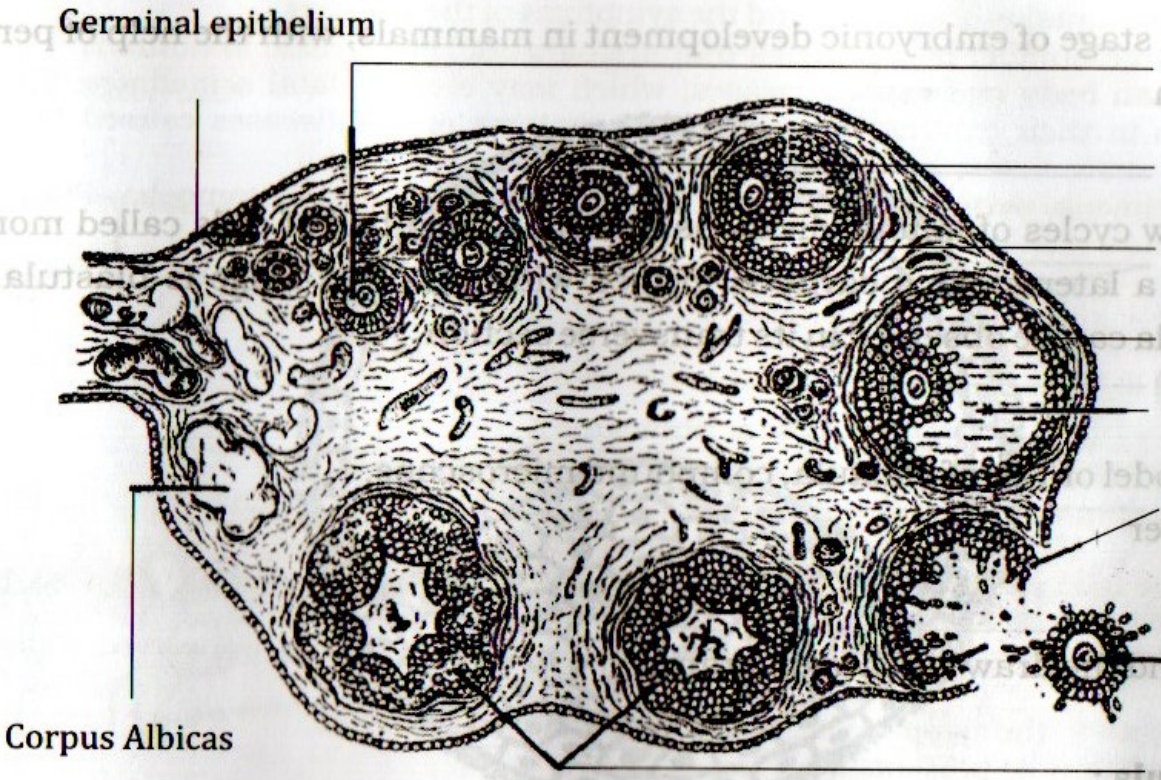
- Each testis is surrounded by a thick, fibrous, vascularised connective tissue capsule called tunica albuginea.
- Each testis is made up of about 250 compartments called testicular labules.
- Each lobule contains 2-3 coiled seminiferous tubules
- Spermatogonia undergo spermatogenesis to produce sperms.
- Supportive and nutritive Sertoli or sustentacular cells.
- Stroma of testis contains intersitial cells called Leydig cells. they secrete testosterone (male sex hormone).



T.S OF MAMMALIAN TESTIS

T.S. of Mammalian Ovary :

- Ovaries represent a pair of female gonads.
- Each ovary is surrounded by a thick, fibrous, capsule called tunica albuginea. This encloses ovarian stroma.
- The wall of ovary is made up of germinal epithelial cells called oogonial cells, They produce ova.
- Matured mammalian follicles are called Graafian follicles.
- After ovulation the ruptured follicle is transformed into a temporary endocrine structure called corpus luteum (yellow body), which secretes progesterone.
- If fertilization fails to occur the corpus luteum degenerates into corpus albicans (white body).
- The granulosa cells of ovary secrete estrogens.



T.S OF MAMMALIAN OVARY

EXERCISE - 14 : BLASTULA OF MAMMALS

Aim: To study the blastula stage of embryonic development in mammals, with the help of permanent slide, chart, model or photograph

Principle:

The zygote undergoes a few cycles of mitotic divisions to form a solid ball of cells called morula. The cells continue to divide and at a later stage a cavity is formed within it. This stage is blastula. The internal structural details of blastula can be observed in its transverse section.

Requirements:

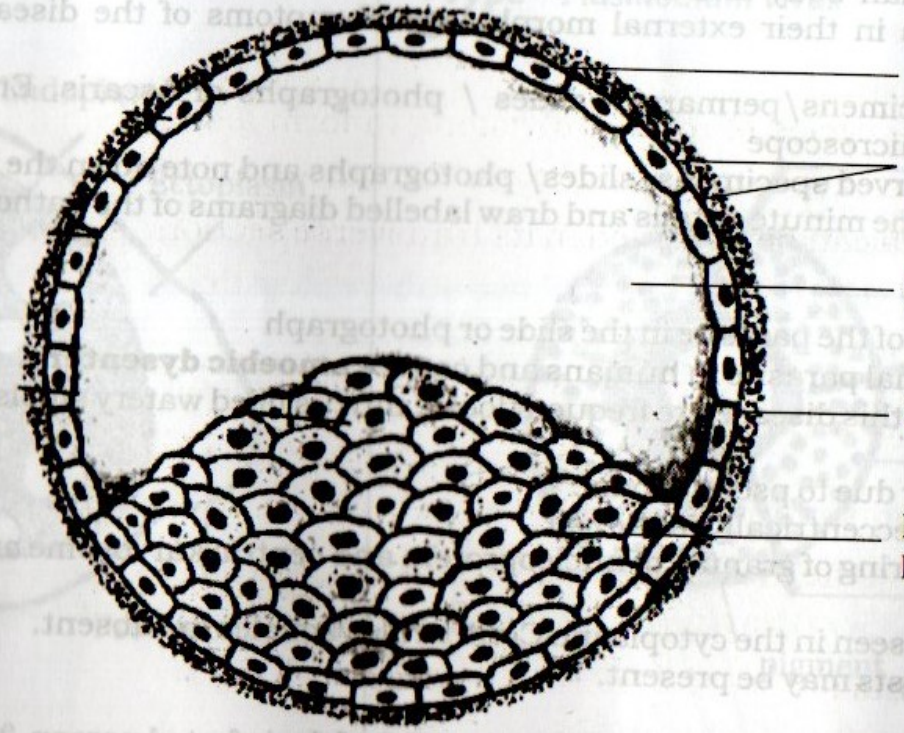
Permanent slide, chart/model of T.S. of blastula, compound microscope,
lens cleaning fluid and paper

Observation :

Record the observations and the draw labelled diagram.

V. S. of Mammalian Blastula :

- Blastula of a mammal is called the blastocyst.
- Blastula is a spherical, hallow embryoinc developmental stage.
- It consists of a covering trophoblast, an inner cell mass and a central flluid filled cavity called blastocel.
- The trophoblast cells which cover over the inner cell mass are called cells of Rauber.
- Trophoblast provides physiological support, it gives rise to extra embryonic membranes and it helps in implantation.



V.S. OF MAMMALIAN BLASTOCYST

EXERCISE - 15 : COMMON DISEASE CAUSING ORANISMS

Aim: To identify common disease-causing organisms and the symptoms of the diseases

Principle: There are quite a large number of organisms that are parasitic/pathogenic to humans. These organisms substantialy damage the human body and cause diseases, which may even be fatal sometimes. These organisms exhibit characteristic features in their external morphology. Symptoms of the diseases caused by them are also specific.

Requirement: Preserved specimens/permanent slides / photographs of Ascaris, Entamoeba, Plasmodium, Ring-worm fungus and compound microscope

Procedure : Observe the preserved specimens/slides/ photographs and note down the features in the practical record book. Take care to observe all the minute details and draw labelled diagrams of the pathogens.

Observation

Entamoeba histolytica

Observe the following features of the parasite in the slide or photograph

Note: Entamoeba is an intestinal parasite in humans and causes **amoebic dysentery.**

Symptoms : The symptoms of this disease are frequent loose, mucus filled watery stools, abdominal pain and spasms.

a. It is unicellular.
b. Shape of the cell is irregular due to pseudopodia.
c. A single nucleus is present eccentrically in the cell.
d. In the nucleus a peripheral ring of granule of nucleoprotein and central karyosome are observed. Rest of the space in the nucleus looks empty
e. A few food vacuoles may be seen in the cytoplasm. Contractile vacuoles are absent.
f. Mature quadrinucleated cysts may be present.

B. *Plasmodium vivax*

a. It is an intracellular endoparasite seen easily within the RBC of the infected person.

Symptoms: Intermittent high fever with chills followed by profuse sweating at an interval of alternate days.

b. It is unicellular.
c. The most diagnostic stage of the parasite is "signet ring" stage in the erythrocytes, within which it appears as a rounded body.
d. It has a big vacuole inside, and the cytoplasm is accumulated at one place containing the nucleus. Because of the above mentioned features, the parasite appears as a ring.

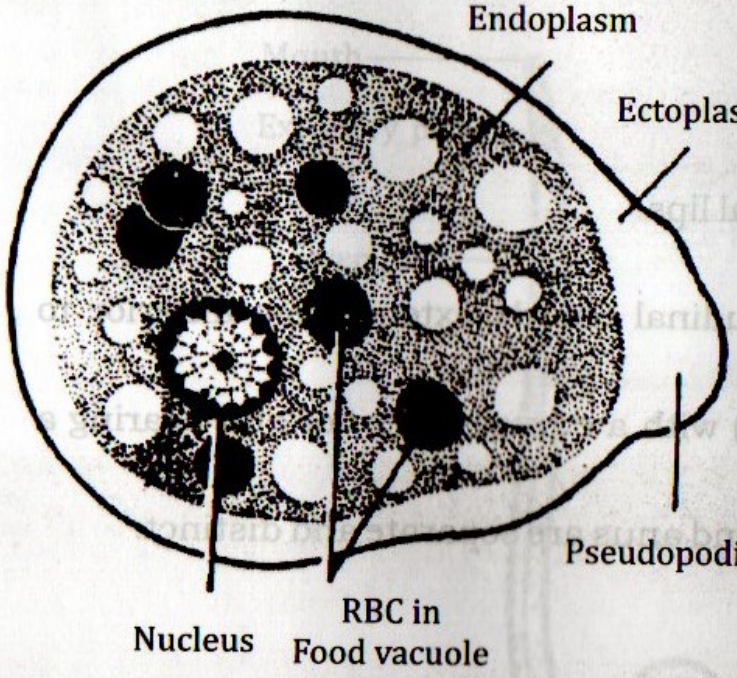
Note: It is a protozoan parasite causing **malaria** in humans. When an infected female anopheles mosquito bites a healthy person, it injects the infective stage, sporozoite, into the peripheral blood vessels. The infective stage undergoes several rounds of multiplication in liver and erythrocytes.

Systematic position

Phylum – Protozoa

Class – Rhizopoda

Type – *Entamoeba histolytica*



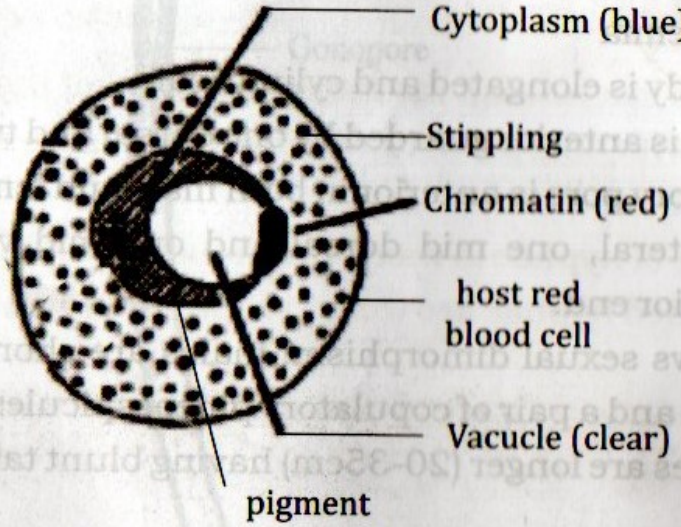
ENTAMOEBA histolytica

Systematic position

Phylum – Protozoa

Class – Sporozoa

Type – *Plasmodium vivax*



PLASMODIUM vivax

***Ascaris lumbricoides* : (Round worm)**

1. It is a monogenetic, endoparasite inhabiting the small intestine of man causing ascariasis.

   Symptoms:

   (a) Irregular bowel,

   (b) Occasional vomiting,

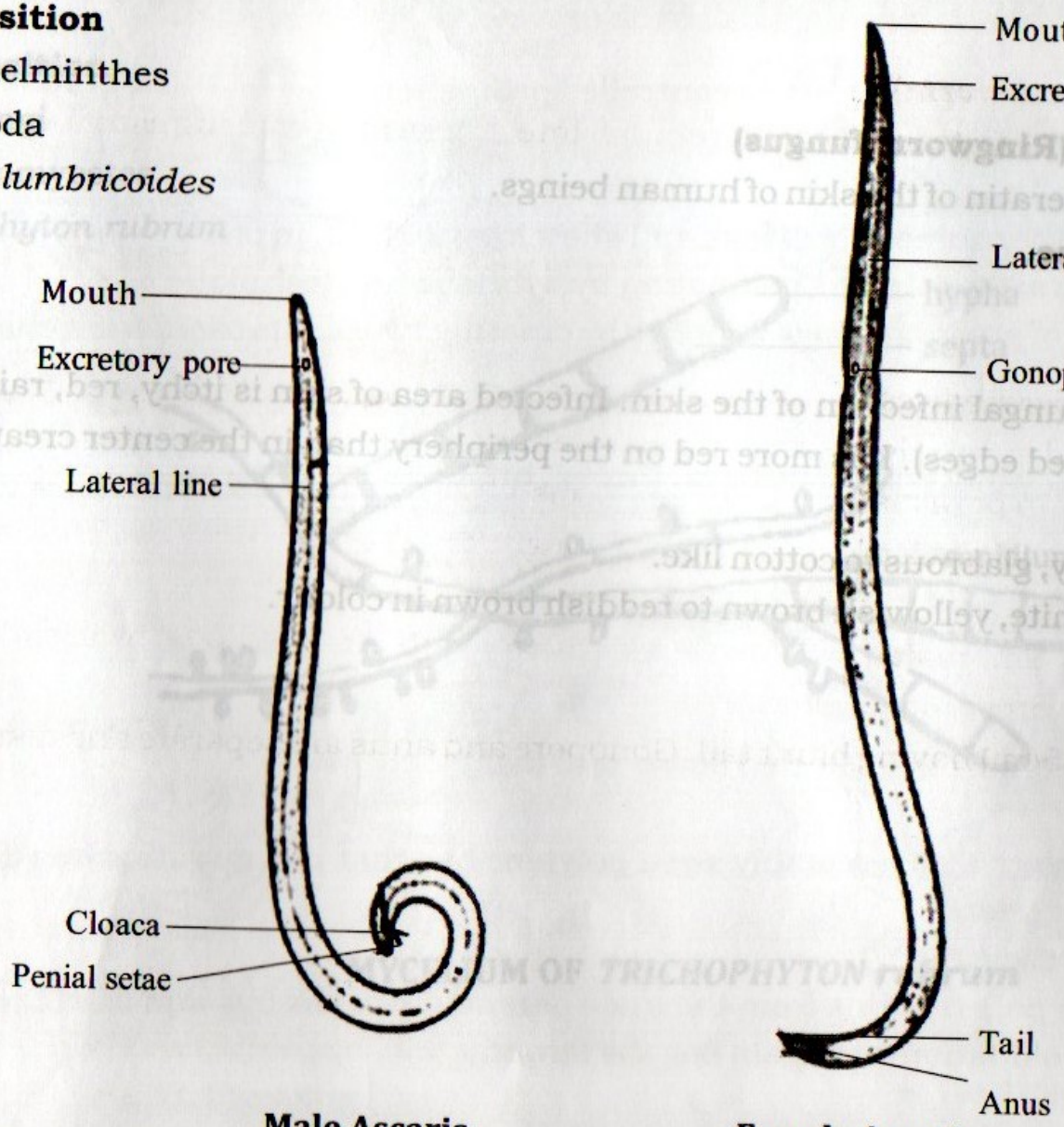
   (c) Anaemia
2. The body is elongated and cylindrical.
3. Mouth is anterior guarded by one dorsal and two sub-ventral lips.
4. Excretory pore is anterior in both male and female worms.
5. Two lateral, one mid dorsal and one mid ventral longitudinal chords extend from anterior to posterior end.
6. It shows sexual dimorphism, males are shorter (15-30cm) with a curved posterior end bearing a cloaca and a pair of copulatory pineal spicules.
7. Females are longer (20-35cm) having blunt tail. Gonopore and anus are separate and distinct.

Systematic position

Phylum – Aschelminthes

Class – Nematoda

Type – *Ascaris lumbricoides*



ASCARIS lumbricoides

D. *Trichophyton rubrum* (Ringworm fungus)

It is a fungus that feeds on keratin of the skin of human beings.

it causes **Ring worm Disease.**

Symptoms

Ringworm is a contagious fungal infection of the skin. Infected area of skin is itchy, red, raised, scaly patches (with sharply defined edges). It is more red on the periphery than in the center creating a ring like appearance.

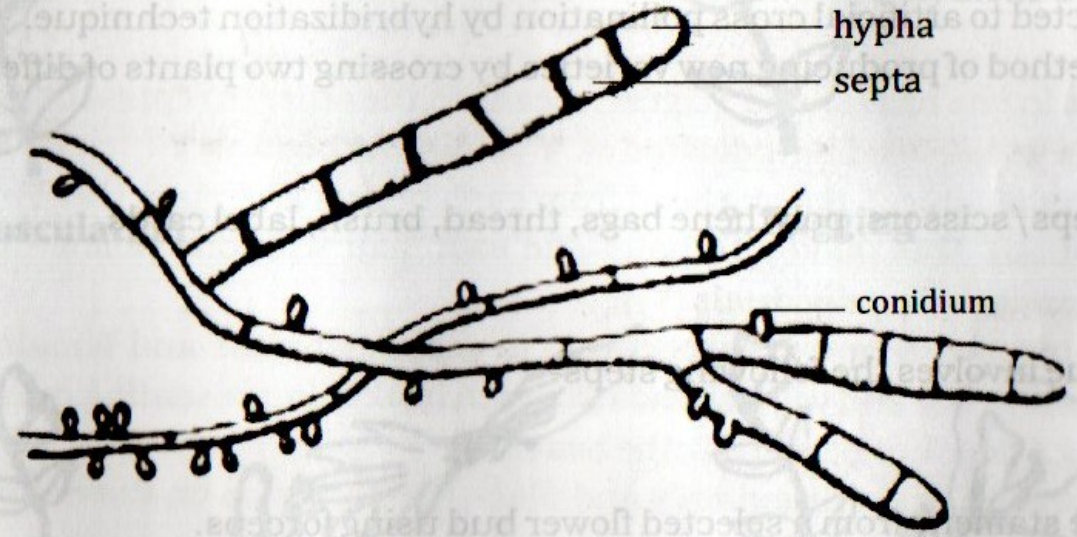
1. Texture of hyphae is waxy, glabrous to cotton like.
2. Unstained hyphae are white, yellowish brown to reddish brown in colour.

Systematic position

Kingdom – Fungi

Class – Deuteromycetes

Type – *Trichophyton rubrum*



MYCELIUM OF *TRICHOPHYTON rubrum*

EXERCISE - 16 : CONTROLLED POLLINATION

Aim : To perform emasculation and to perform a controlled pollination.

Principle :

The self pollinated plants or the inbreeds are weak and show low yield. So, in order to improve their quality, the inbreeds are subjected to artificial cross pollination by hybridization technique.
Hybridization is the method of producing new varieties by crossing two plants of different characters.

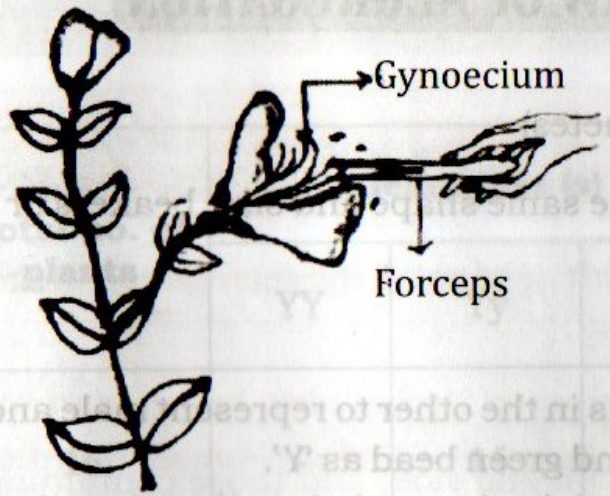
Requirements :

Flowering plants, forceps/scissors, polythene bags, thread, brush, label cards.

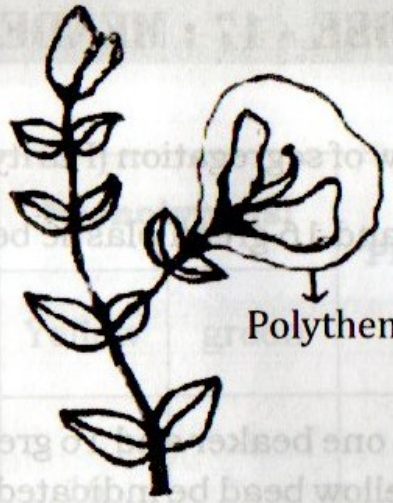
Procedure :

Hybridization technique involves the following steps-

1. **Emasculation :**
   Carefully remove the stamens from a selected flower bud using forceps.
2. **Bagging :**
   Cover the emasculated flower with a suitably sized polytene bag that has minute pores (for aeration) to prevent undesirable pollination.
3. **Collection of pollen and crossing :**
   Collect mature and viable pollen grains from the male parent. Open the bag and dust the pollen on the stigma to achieve cross-pollination and again bag the flower.
4. **Tagging :**
   Tag the plants by writing important information like date and name of the seed parent.



1. Emasculation



2. Bagging

3. Crossing

4. Tagging

CONTROLLED POLLINATION

EXERCISE - 17 : MENDEL'S LAW OF SEGREGATION

Aim: To verify Mendel's law of segregation (Purity of Gametes)

Requirement: 16yellow and 16 green plastic beds of the same shape and size, beakers or petri dishes, and a napkin or hand towel.

Procedure:

1. Put 16 yellow beads in one beaker and 16 green beads in the other to represent male and female gametes respectively. Let the yellow bead be indicated as 'Y' and green bead as 'Y'.
2. Spread a napkin on the table. Take one bead from each beaker and place them in pairs on the napkin (it represent fertilization). This makes 16 pairs and each pair with one yellow - Y and one green bead-y representing heterozygous F_1individuals (Yy).
3. Take two beakers labelled A and B respectively to represent male and female individuals. Put 8 F_1 progeny i.e., 8 pairs of green and yellow beads in easch beaker (representing maless and females).
4. Stir the beads using a pencil or pen to mix the beads.
5. Take one bead from each beaker randomly and place them in pairs on the napkin. This makes 16pairs representing 16 F_2individuals.
6. Note the genotype (YY or Yy or yy) of each pair, and their possible phenotype (keep in mind that yellow is dominant over green).
7. Determine the genotypic and phenotypic ratios of the pooled data.

OBSERVATION: Record the results in the following table.

Generation	Total no. of plants	Genotypes (s)			Phenotype (s)		Phenotypic ratio	Genotypic ratio
		YY	Yy	yy	Yellow	green		
F_1	16	-	16	-	16	-	-	-
F_2	16	3	10	3	13	3	4:3:1	1:3:1
F_3	16	3	10	3	13	3	4:3:1	1:3:1
F_4	16	5	6	5	11	5	2:4:1	1:3:1

EXERCISE - 18 : MENDEL'S LAW OF INDEPENDENT ASSORTMENT

Aim: To verify Law of independent assortment

Principle:

In a dihybrid cross, the segregation of one gene pair is independent of the segregation of the other pair.

Procedure:

1. Put 16 beads of each colour in four separate beakers.
2. Keep the beakers containing yellow and red beads on one side, and those containing green and white beads on another side. They represent pure line parents bearing yellow seed with red flower (dominant traits - YYRR) and green seed with white flower (recessive traits -yyrr). The yellow and green bead represent two allets of a gene for flower colour.

DOMINANT PARENT		recessive parent	
BEAKER-01 YELLOW BEADS	BEAKER-02 RED BEADS	beaker - 03 green beads	beaker - 04 white beads

3. Take one yellow, one green, one red and one white bead and place them together on the napkin spread on the table. . Repeat the same till all the beads are utilized. **Note:** each cluster containing one yellow, one green, one red and one white bead represent F_1 individuals (YyRr).
4. Take four beakers and label them as seed colour in male, flowe colour in male seed colour in female and flower colour in female respectively.

F_1 - MALE		F_1 - FEMALE	
BEAKER-01 YELLOW & GREEN BEADS	**BEAKER-02** RED & WHITE BEADS	**BEAKER-03** RED & GREEN BEADS	**BEAKER-04** RED & WHITE BEADS

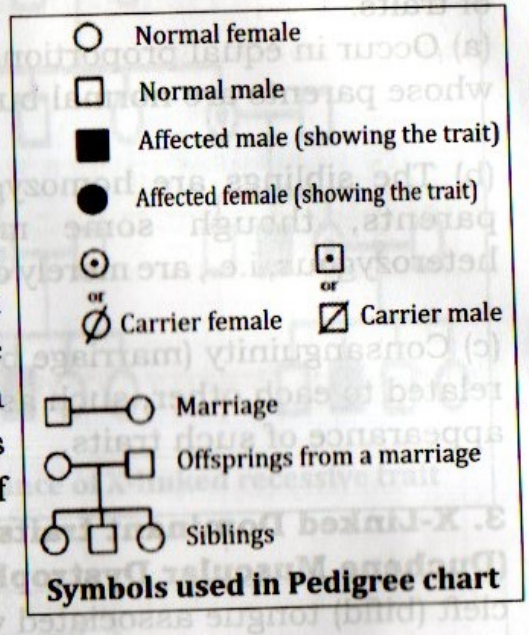
5. In the beakers labelled as seed colour in male put yellow and green beads from the first 8 clusters and similarly in the beakers labelled as seed colour in female put yellow and green beads from the other 8 clusters.
6. In the same way in the beaker labelled as flower colour in male put red and white beads froms the first 8 clusters and similarly in the beakers represent F_1 male and female individuals.
7. Stir the beads of each beaker using a pencil o pen to mix the beads
8. Take one bead from each beaker randomly and place them together on the napkin. Repeat this same till the last bead is utilized. This makes 16 clusters each having 4 beads representing 16 F_2 individuals.
9. Record the genotypic and phenotype of each of the 16 F_2 individuals.
10. Repeat the experiment a few more time and tabulate the result Determine the phenotypic ratio.

Generation	Trials	Trial No. of plants	Genotypes									Phenotypes			
			RRYY	RrYY	RRYy	RrYy	rrYY	rrYy	RRyy	Rryy	rryy	RED YELLOW	whiteYELLOW	RED green	white green
F_1	1	16	-	-	-	16	-	-	-	-	-	16	-	-	-
	2	16	-	-	-	16	-	-	-	-	-	16	-	-	-
	3	16	-	-	-	16	-	-	-	-	-	16	-	-	-
Total															
F_2	1	16	2	1	1	6	1	1	1	1	2	10	2	2	2
	2	16	-	3	1	8	-	1	-	3	-	12	1	3	0
	3	16	1	3	1	5	-	2	-	4	-	10	2	4	0
Total			3	7	3	19	1	4	1	8	2	32	5	9	2

EXERCISE - 19 : PEDIGREE CHARTS

Aim : Preparation and analysis of Pedigree Charts

Principle : The Mendelian concept of dominance and segregation can also be studied in humans by preparing and then analyzing the pedigree charts. The internationally approved symbols are given below.



Symbols used in Pedigree chart

PROCEDURE : Select a family in which any one of the monogenic traits such as tongue rolling, widow's peak, blood groups', red-green colour blindness, dimple in the cheek, hypertrichosis of ear, hitch-hiker's thumb, etc., is found. Ask the person exhibiting the trait to tell in which of his/her parents, grand parents (both maternal and paternal), their children and grand children the trait in question is present. The information made available is the basis for the preparation of pedigree chart using the appropriate symbols.

1. Autosome Linked Dominant traits :

{Brachydactyly(shortening of the fingers and toes due to unusually short bones), polydactyly (a condition in which a person has more than five fingers per hand or five toes per foot), dimple in the cheek, e.t.c.,}

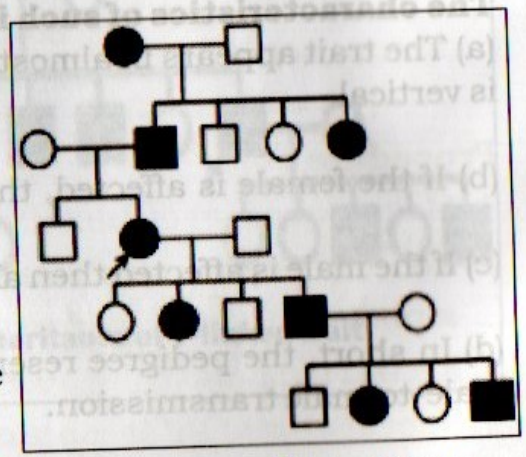
The characteristic features of inheritance of such type of traits are :

(a) Transmission of traits occurs from parents of either sex.

(b) Males and females are equally affected.

(c) The pedigree is vertical, i.e., the trait is marked to be present in each of the generations.

(d) Multiple generations are characteristically affected.



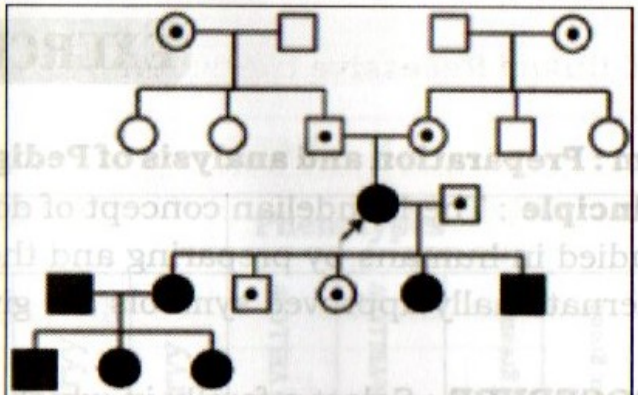
Autosomal linked traits

2. Autosomal Recessive trait : {Eg., Albinism}

The following are the salient features of the inheritance of such type of traits.

(a) Occur in equal proportions in multiple male and female siblings, whose parents are normal but carriers;

(b) The siblings are homozygous for the defective allele, but their parents, though some may appear normal, are obviously heterozygous, i.e., are merely carriers of the trait.



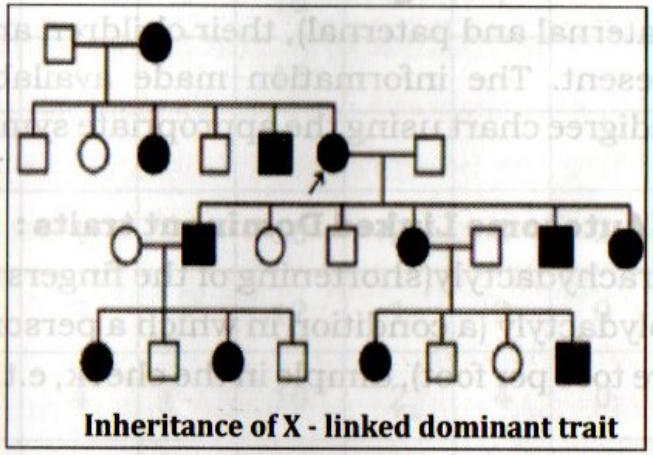
Inheritance of autosomal linked recessive trait

(c) Consanguinity (marriage between man and woman genetically related to each other, such as cousins) occasionally results in the appearance of such traits.

3. X-Linked Dominant traits:{ E.g., oral-facial-digital syndrome **(Duchene Muscular Dystrophy),** which results in absence of teeth, cleft (bifid) tongue associated with mental retardation}

The characteristics of such inheritance are:

(a) The trait appears in almost all the generations, and the inheritance is vertical.



Inheritance of X - linked dominant trait

(b) If the female is affected, then about half of her sons are affected.

(c) If the male is affected then all of his daughters would be affected, but none of his sons are affected.

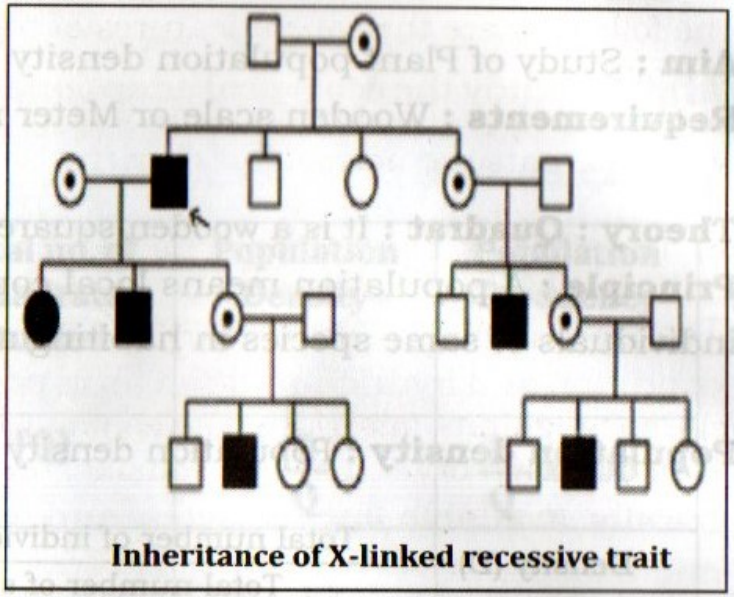
(d) In short, the pedigree resembles the pattern of inheritance of autosomal dominants, except that there is no male-to-male transmission.

4. X-linked Recessive traits: {E.g., Red-green colour blindness and hemophilia}

(a) Females express the trait only when they are homozygous for the mutant allele, whereas the males do so even when they are hemizygous for it.

(b) About half of the sons of the carrier (heterozygous for the trait) females are affected. In case of homozygous females showing the trait, fifty percent of her daughters and all of her sons are likely to be affected. Therefore, the males are most affected in the population.

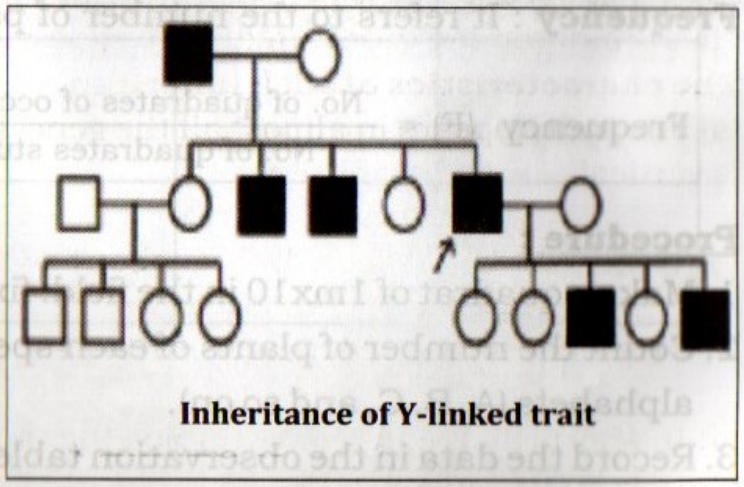
(c) Affected persons are related to one another through the maternal side of their family. Any evidence of male-to-male transmission of the trait rules out the X- linked inheritance.



Inheritance of X-linked recessive trait

5. Y-chromosome linked traits : {E.g. Hypertrichosis of the ear (presence of hairs on pinna)}

These are the traits whose gene is present on the Y-chromosome. The females do not have any Y-chromosome, whereas all the males must have a Y-chromosome to be a male, and this Y-chromosome they get from their father. Therefore, any trait linked to the Y- chromosome must be present only in males, and certainly not in any of the females. This is why these traits are also called male-sex limited traits. All the sons of the affected male would express the trait whereas none of his daughters would do so.



Inheritance of Y-linked trait

EXERCISE - 20 : PLANT POPULATION DENSITY AND FREQUENCY

Aim : Study of Plant population density by quadrat method.

Requirements : Wooden scale or Meter rod, Pegs, Rope, Practical file etc.

Theory : Quadrat : It is a wooden square (50 cm x 50 cm) which has small squares (approx 10 cm x 10 xm)

Principle : A population means local community of physiologically or morphologically similar individuals of same species in habiting a particular area.

Population density : Population density means number of individuals present in given area at given time.

$$\text{Density (D)} = \frac{\text{Total number of individuals in all quadrates}}{\text{Total number of studied quadrates}}$$

Frequency : It refers to the number of particular species in a total quadrates at given time.

$$\text{Frequency (F)} = \frac{\text{No. of quadrates of occurrence}}{\text{No. of quadrates studied}}$$

Procedure :

1. Make a quadrat of 1mx10 in the field. fix four nails at the corners of quadrat and tie thread over the nails.
2. Count the number of plants of each species present in the quadrat. Indicate the plants by their names or alphabets (A, B, C, and so on).
3. Record the data in the observation table.
4. Repeat atleast three more quadrats and record the data.

Plant species	No. of Individuals per Quadrat				Total no. of Individuals of all Quadrate	Total no. of Quadrats in which species occurred	Total no. of Quadrats Studied	Population Density	Population Frequency
	1	2	3	4	(S)	(N)	(Q)	$\frac{S}{Q}$	$\frac{N}{Q} \times 100$
A	6	0	2	2	10	3	4	2.5	75
B									
C									

EXERCISE - 21 : ECOLOGICAL ADAPTATIONS OF XERIC AND HYDRIC PLANTS

Aim: To study the ecological adaptions of plants living in xeric

Principle:

Adaptation is successful adjustment of organism to prevailing environment conditions. Plants exhibit very high level of adaptations (plasticity) due to their inability to migrate the place where they live. Terrestrial plants are found in three types of habitat namely xeric condition as in deserts, mesic condition and hydric condition as in ponds, rivers, oceans etc.

Requirements :

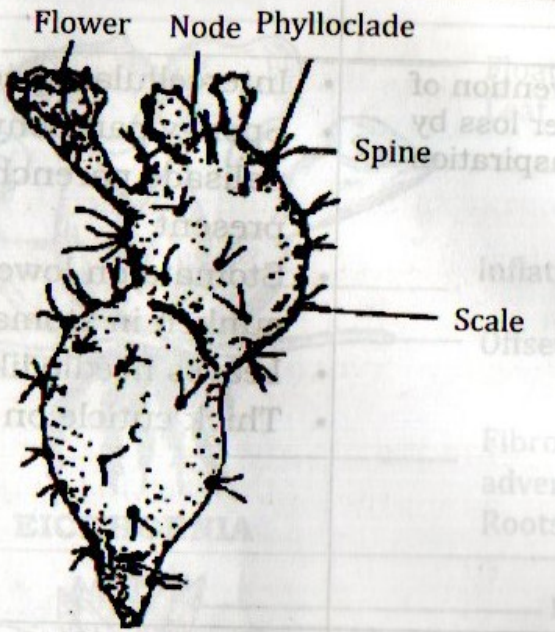
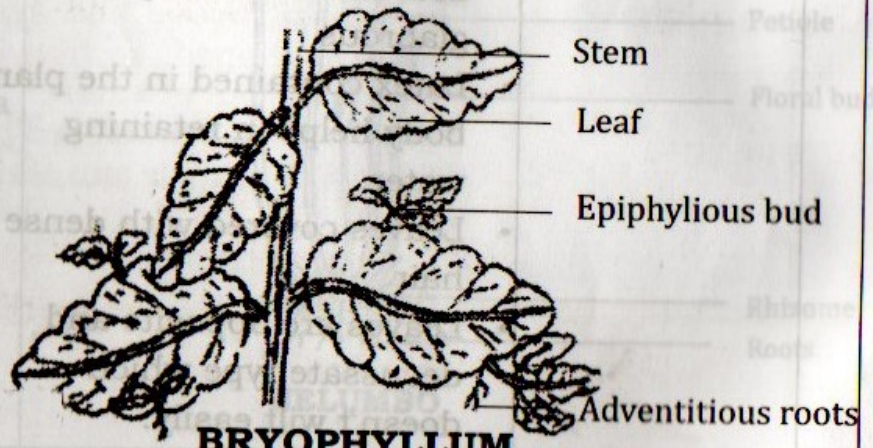
Locally available hydrophytes and xerophytes, beakers, glassjars, slides, cover slips, blade and microscope.

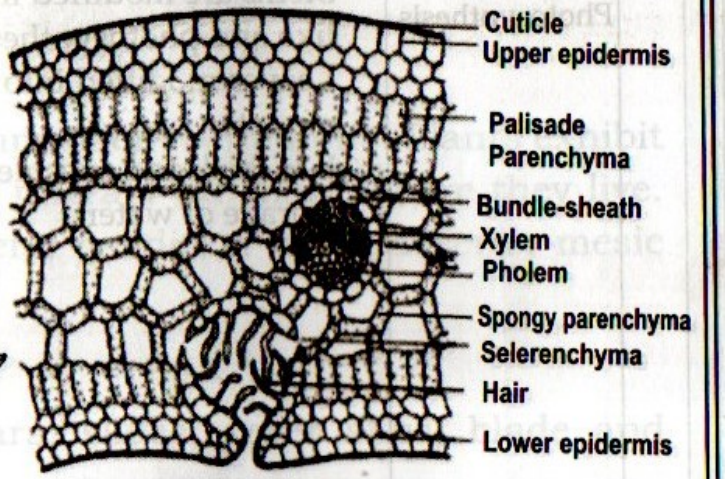
Xerophytes : Opuntia, Euphorbia, Bryophyllum, Acacia (Babool), Calotropis, Casuarinas, Nerium, Parkinsonia, Cycas leaflets, Pine needles etc.

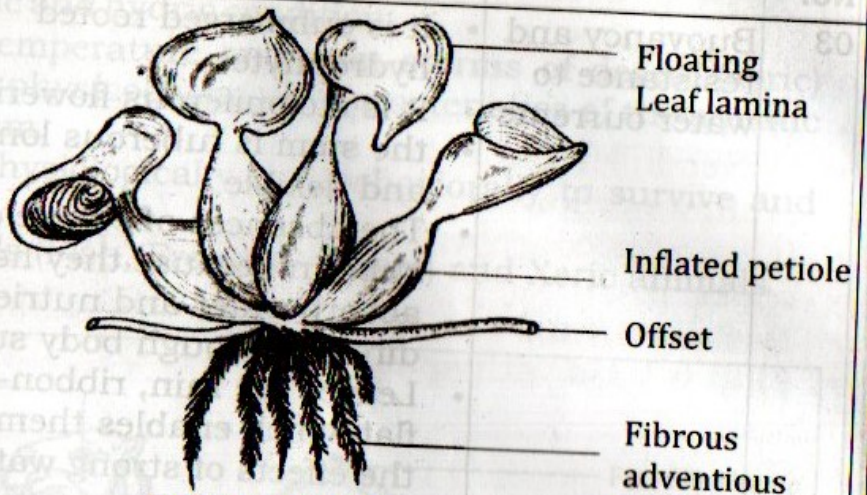
Hydrophytes : Hydrilla, Vallisneria, Pistia, Jussieua, Eichhornia, Salvinia, Utricularia, Rhizophora etc.

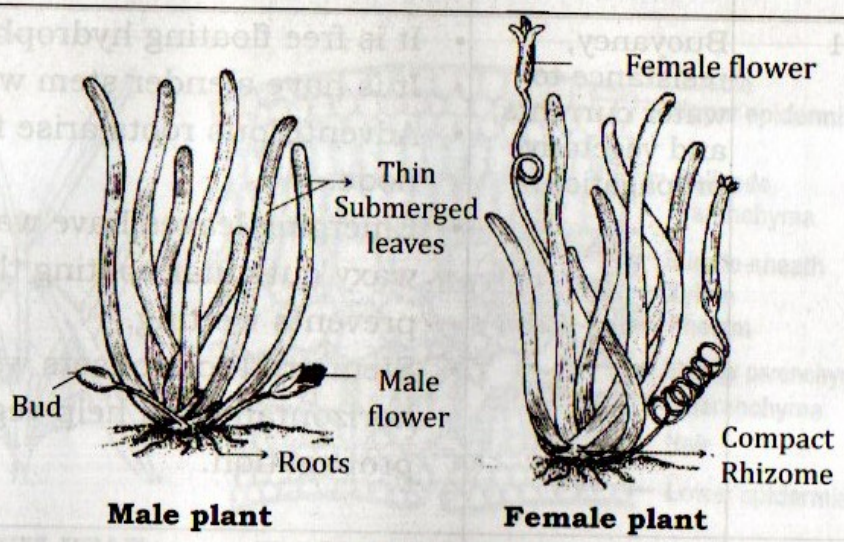
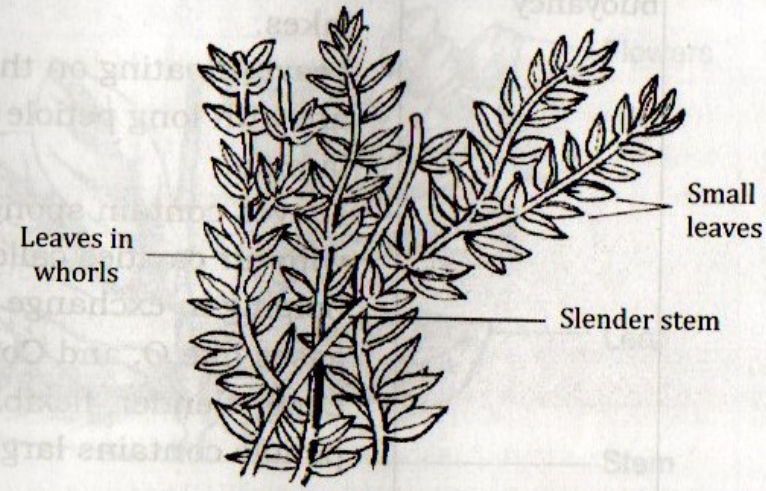
Procedure:

1. Collect the locally available hydric and xeric plants.
2. Prepare a temporary stained T/S of leaf, stem and root of collected plant specimen.
3. Study their morphological and anatomical adaptations.
4. Write the name of the plant in which that particular adaptation is observed.

Sl. No.	Adaptations	Modifications (Morphological or Anatomical)	Examples
01.	Conservation of water & Photosynthesis	• It shows phylloclade modification. • Stems are modified into flattened green like and photosynthetic in nature. • Leaves modified into spines to reduce loss of water. • It is thick and succulent with mucilage for storage of water	 OPUNTIA
02	Storage of water & Vegetative propagation	• Leaf is modified, thick, leathery, fleshy and filled with succulent tissue to store water. • Leaf margins produce adventitious (epiphyllous) buds which develop adventitious roots. • Leaf buds later develop into daughter plants helping in vegetative propagation. • Presence of Sunken stomata at lower epiderms. • It shows CAM cycle for prevention of Transpiration.	 BRYOPHYLLUM

Sl. No.	Adaptations	Modifications (Morphological or Anatomical)	Examples
03	Prevention of water loss by transpiration	• Intercellular spaces reduced • Spongy parenchyma/ palisade parenchyma present • Stomata on lower surface, sunken in stomatal pits • Leaves needle like • Thick cuticle on leaf surface	• NERIUM LEAF  T.S. OF NERIUM LEAF
04	Prevention of excessive heat	• Leaves are covered by thick cuticle • Leaf surfaces shiny and glabrous • Latex contained in the plant body helps in retaining water. • Leaves covered with dense hair • Leaves are opposite and decussate type which doesn't wilt easily.	• CALOTROPIS

Sl. No.	Adaptations	Modifications (Morphological or Anatomical)	Examples
01	Buoyancy, resistance to water currents, and vegetative propagation.	• It is free floating hydrophyte • It is have slender stem with nodes. • Adventitious roots arise from the node. • Emerging leaves have waterproof, waxy cuticular coating that prevents wetting. • Stem produces offsets which grow horizontally and help vegetative propagation.	 **EICHHORNIA**
02	Exchange of gases & buoyancy	• It is floating rooted hydrophyte commonly found in ponds and lakes. • Leaves floating on the water surface are with long petiole and broad lamina. • Leaves contain spongy parenchyma with air cavities called aerenchyma helping in exchange and storage of gases like O_2 and Co_2. • Long, slender, flexible and spongy petiole contains large air cavities.	**NELUMBO**

Sl. No.	Adaptations	Modifications (Morphological or Anatomical)	Examples
03	Buoyancy and resistance to water currents	• It is submerged rooted hydrophytes • It is stoloniferous flowering plant • the stem is tuberous long slender and flexible • The absences of cuticle or suberin because they have no absorb water and nutrients directly through body surface. • Leaves are thin, ribbon-like and flat which enables them to escape the effects of strong water currents.	 **Male plant** **Female plant** **VALLISNERIA**
04	Prevent rotting effect of water	• The stem is soft and slender and bears thin membranous leaves in whorls of 3-8. • Mechanical tissue and xylem are poorly developed, so that it limps when taken out of water. • Xylem is poorly developed because of lacking of cuticle and stomata • The whole plant is covered by mucilage to prevent rotting effect of water	 **HYDRILLA**

EXERCISE - 22 : ECOLOGICAL ADAPTATIONS OF XERIC AND HYDRIC ANIMALS

Aim :To study the ecological adaptation of animals living in xeric and hydric condition.

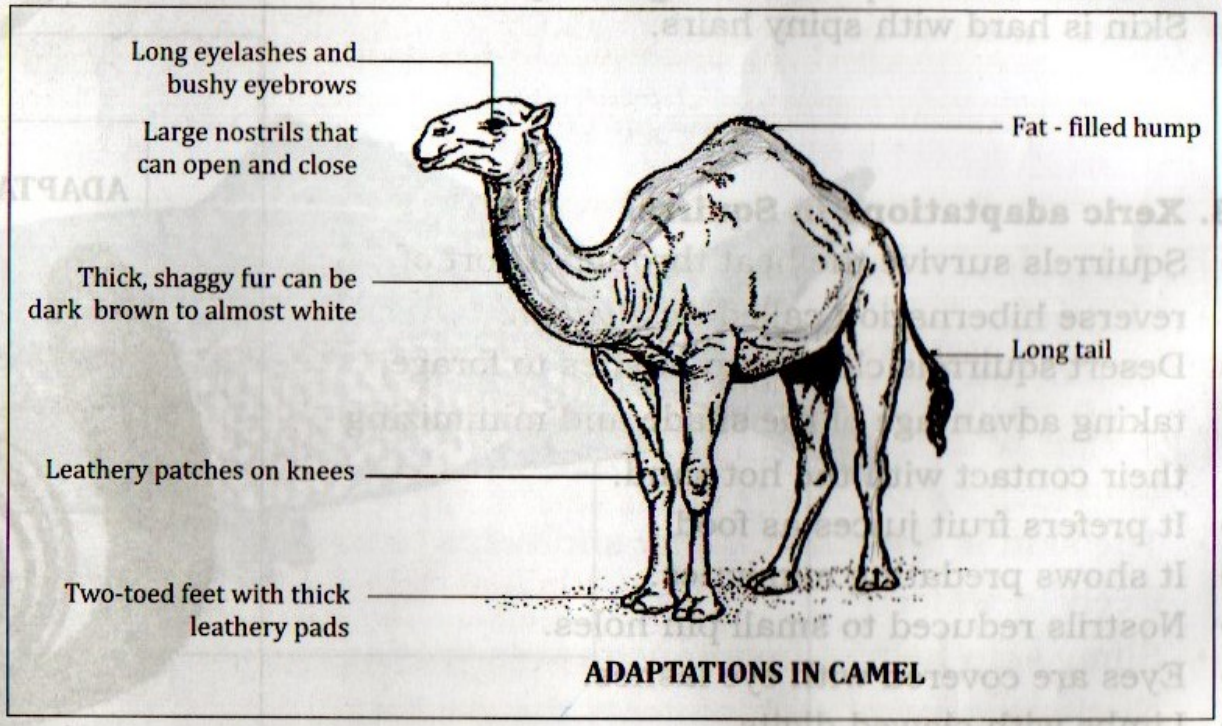
Principle: the factors such as scarcity of water. extreme temperature and dust storms of desert (xeric) ecosystem; temperature of water, penetration of sunlight, the physicochemical characteristics of the aquatic (hydric) ecosystem affect tne growth and survival of the organism.

The organism are designed morphologically, anatomically, physiologically and behaviorally to survive and thrive in their habitats.

Requirement: Specimens/ models/ pictures of hydric animals (Fish, Frog and Prawn) and Xeric animals (Rat, Camel and Squirrel.

Xeric adaptations in Camel :

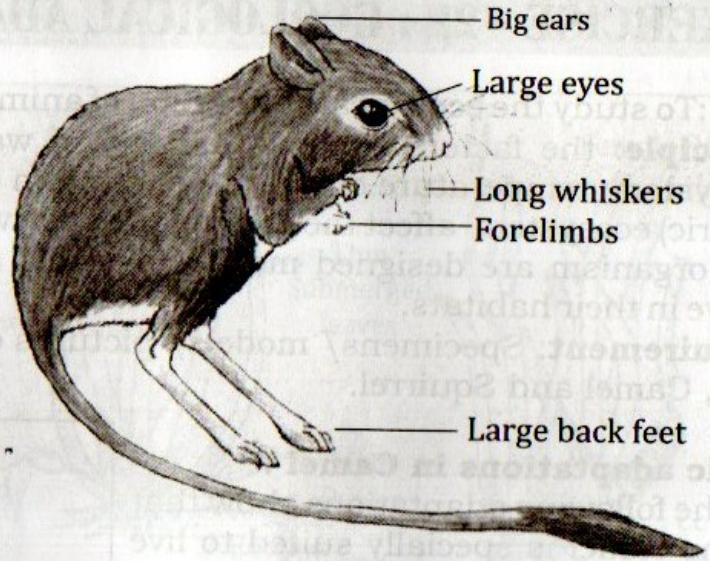
- The following adaptations show that the camel is specially suited to live in the desert.
- Two rows of long eyelashes protect the eye against blowing sand and the sun
- Fat stored in hump helps it to survive long periods without food and water
- Limbs are long and with padded foot.
- Thick leathery patches on knees protect it from getting burn when it kneels on the hot desert sand.
- Broad, flat, leathery pads at the bottom of their hooves spread out when the camel places its feet on the ground thus creating a "snowshoe effect" and preventing the camel from sinking into the sand
- It contains Valvular nostrils.



ADAPTATIONS IN CAMEL

2. Xeric adaptations in Kangaroo rat

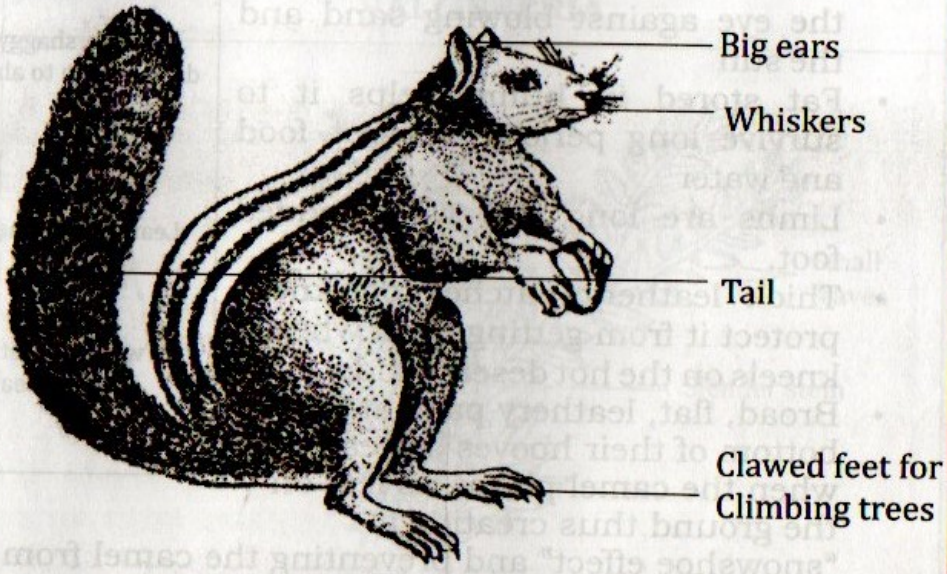
- The kangaroo rat is almost perfectly adapted to life in the desert.
- They can survive without ever drinking any water.
- It lives in Humid holes.
- It shows protecting mimicry.
- It excretes concentrated urine to avoid loss of water.
- Nostrils turned upward in high temperature.
- Skin is hard with spiny hairs.



ADAPTATIONS IN KANGAROO RAT

3. Xeric adaptations in Squirrel

- Squirrels survive the heat through a sort of reverse hibernation called *aestivation.*
- Desert squirrels climb into bushes to forage, taking advantage of the shade and minimizing their contact with the hot sand.
- It prefers fruit juices as food
- It shows predating mimicries.
- Nostrils reduced to small pin holes.
- Eyes are covered with eye lashes.
- Limbs with clawed digits.
-

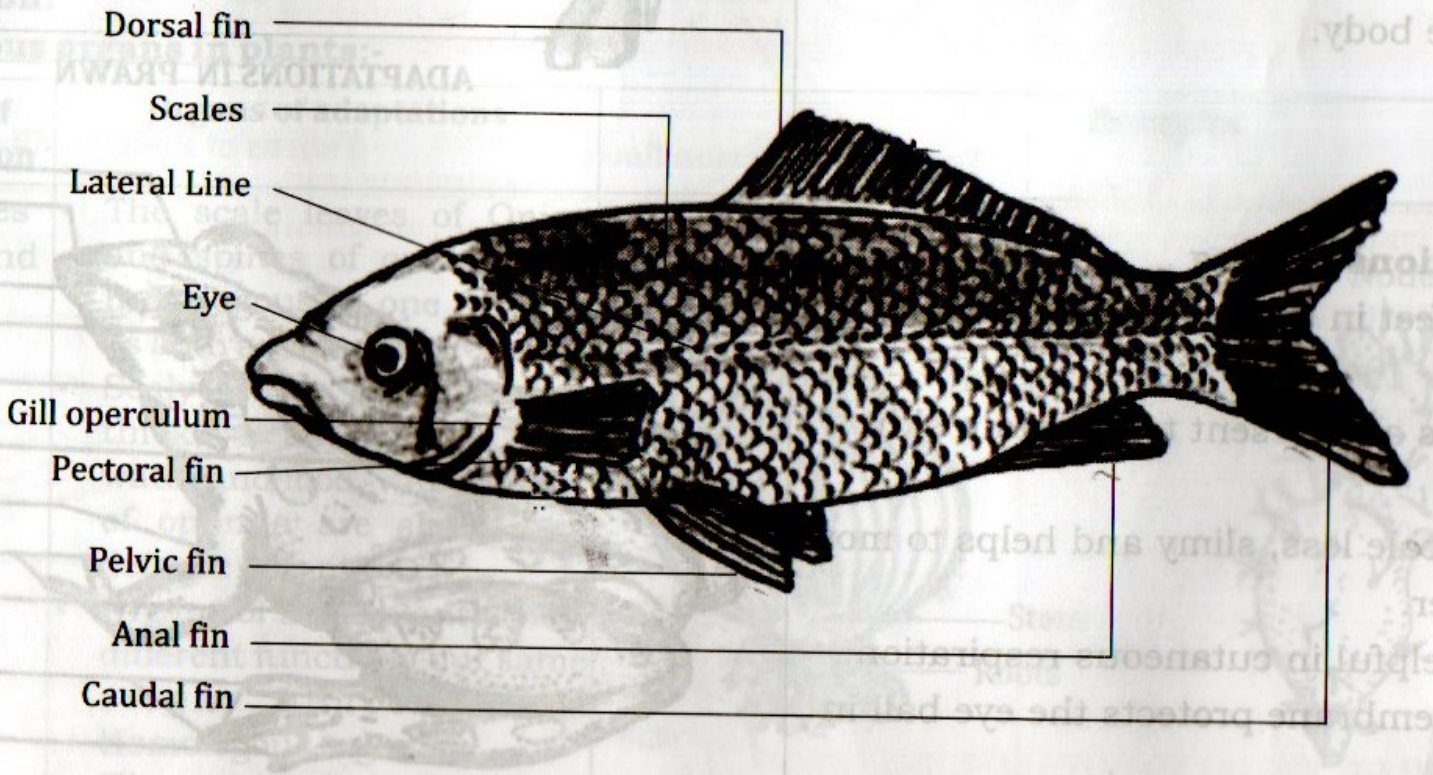


ADAPTATIONS IN SQUIRREL

HYDRIC ANIMALS:

Hydric adaptations in fish

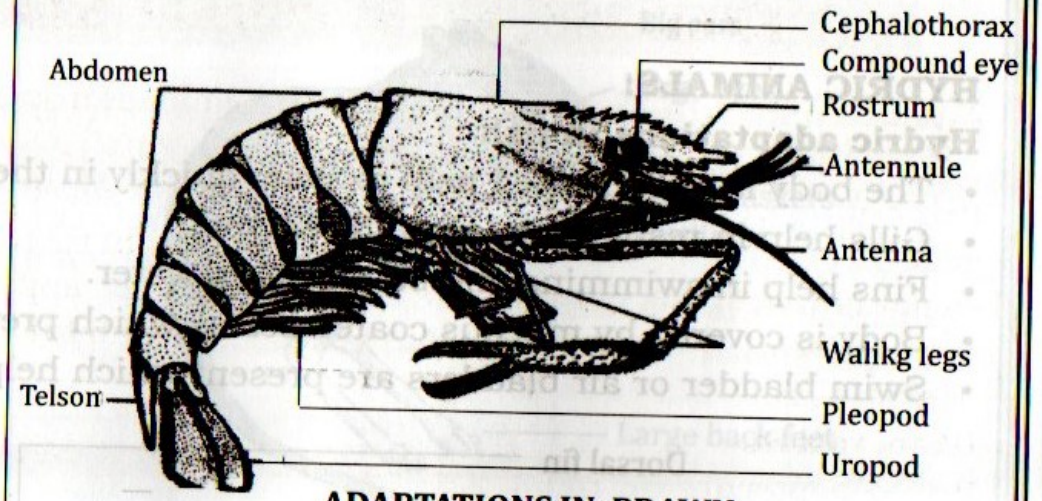
- The body is stream lined help to swim quickly in the water.
- Gills help in respiration inside the water
- Fins help in swimming and balancing in water.
- Body is covered by mucous coated scales which prevent the entry of water into the body.
- Swim bladder or air bladders are present which helps to flot in water.



ADAPTATIONS IN FISH

Hydric adaptations in Prawn

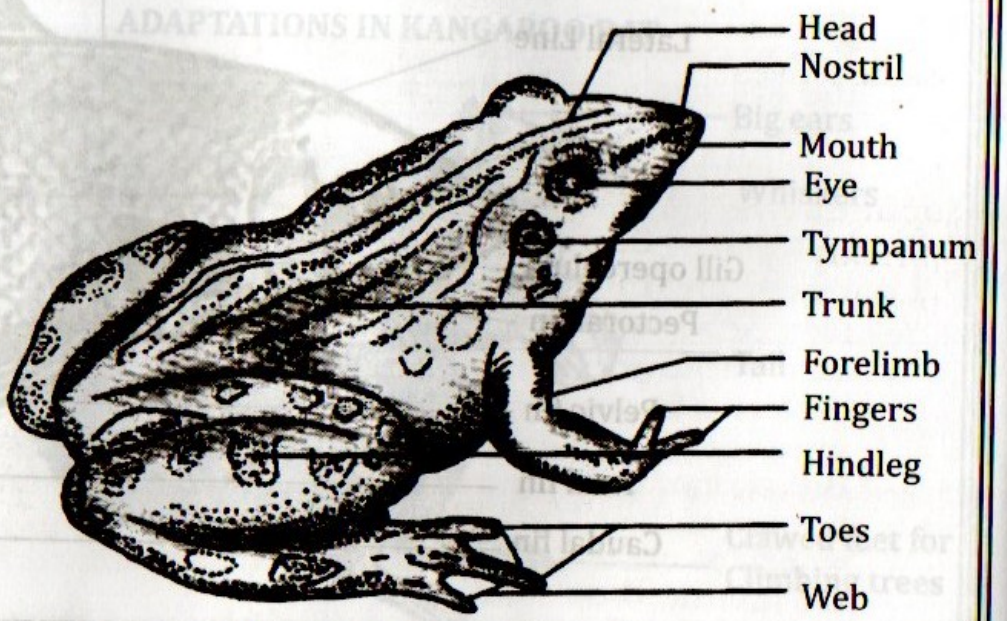
- The stream lined body helps in swimming.
- The pleopods/ swimmeret are well developed for swimming.
- Tail is enlarged helps to balance the body while swimming.
- Chitinous ecoskeleton prevents the entry of water into the body.



ADAPTATIONS IN PRAWN

Hydric adaptations in Frog

- The webbed feet in swimming efficiently in water
- Mucus glands are present to maintain skin moist
- The skin is scale less, slimy and helps to move freely in water.
- Slimy skin helpful in cutaneous respiration.
- Nictitating membrane protects the eye ball in water.



ADAPTATIONS IN FROG

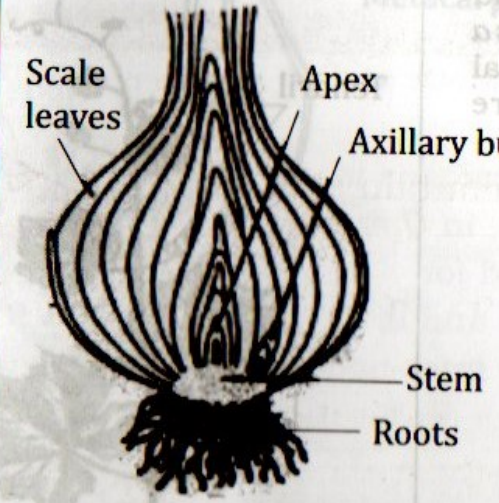
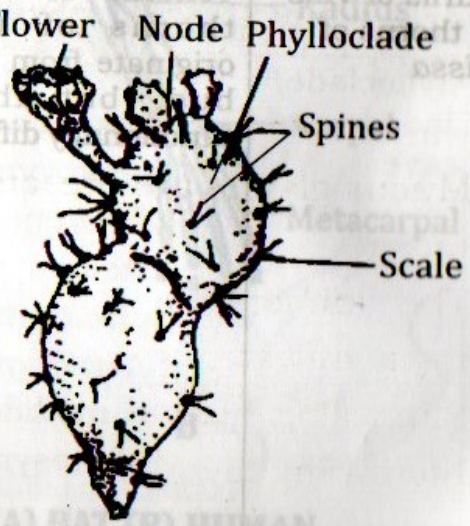
EXERCISE - 23 : HOMOLOGOUS ORGANS IN PLANTS & ANIMALS

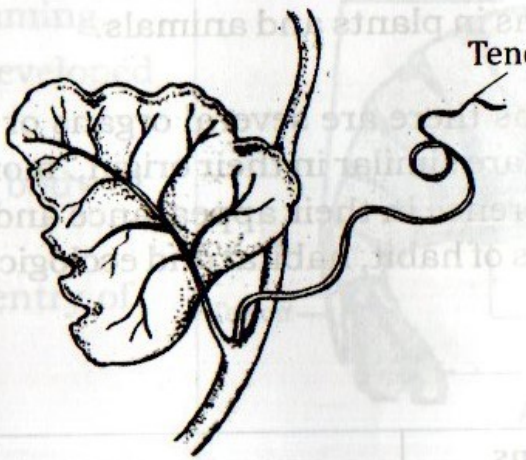
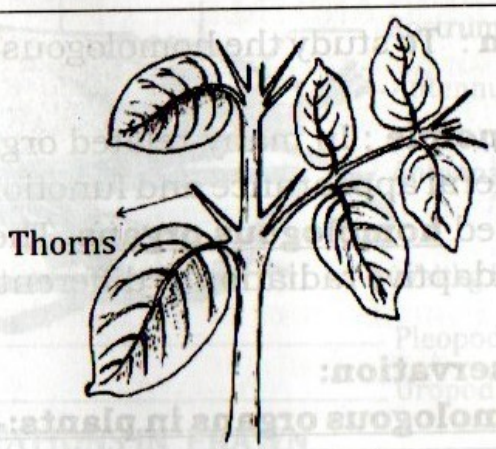
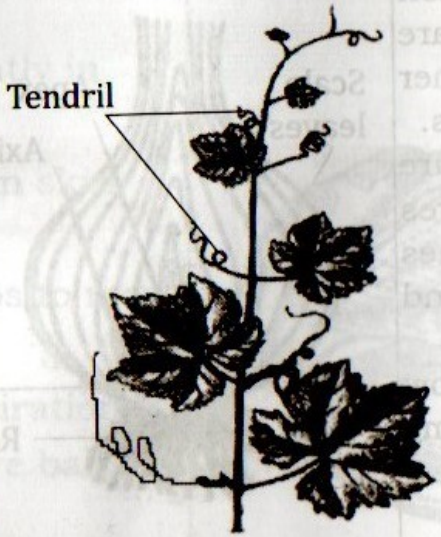
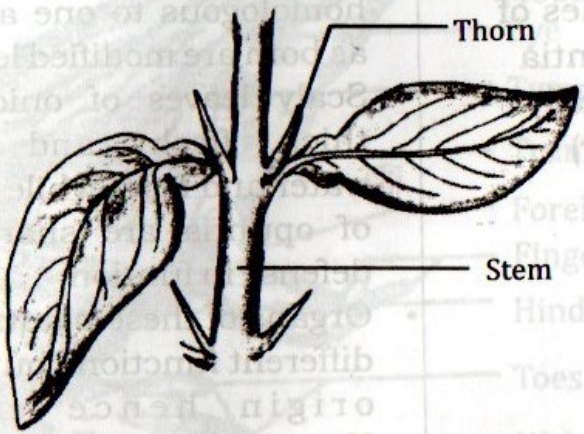
Aim : To study the homologous organs in plants and animals.

Principle : In many related organisms there are several organs or structure showing dissimilarity in their general appearance and function but are similar in their origin,. morphology and anatomy,. Such organs are called **homologous organs.** The difference in their appearance and function are a result of divergence, due to adaptive radiation to different kinds of habit, habitat and ecological niche.

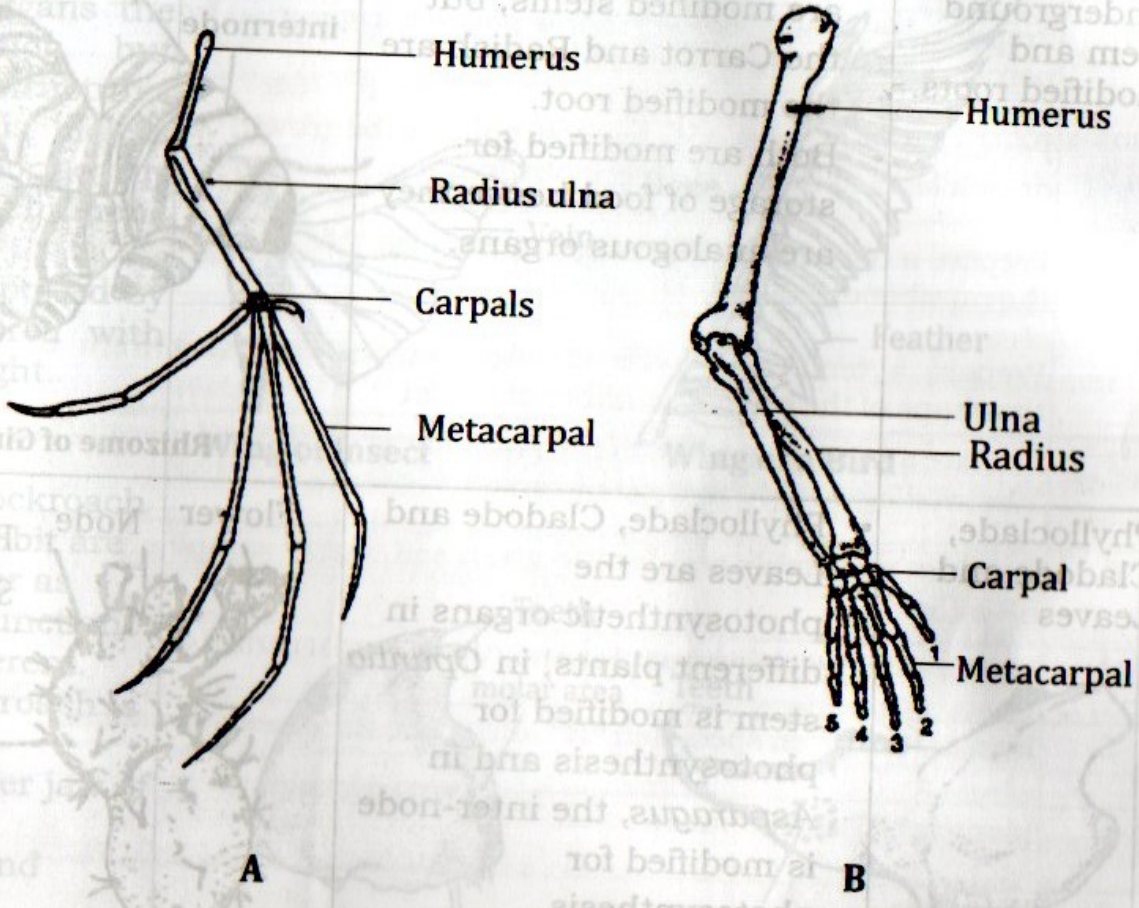
Observation:

Homologous organs in plants:-

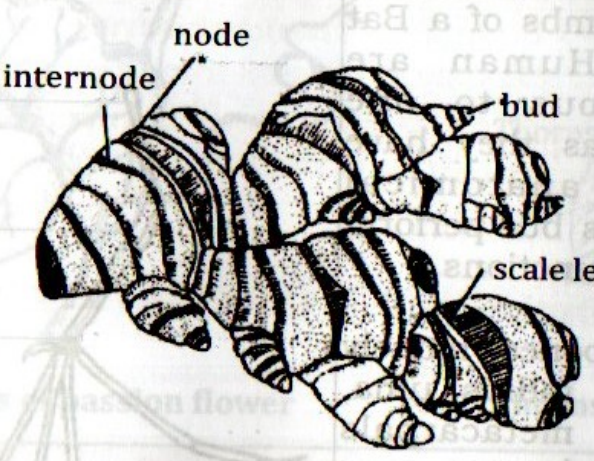
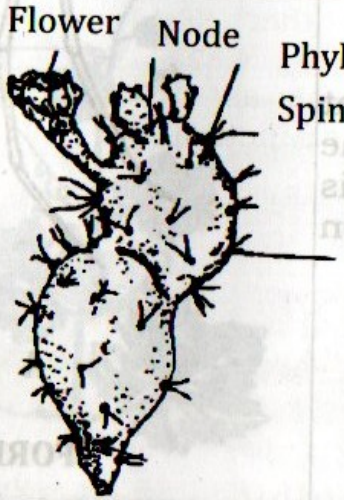
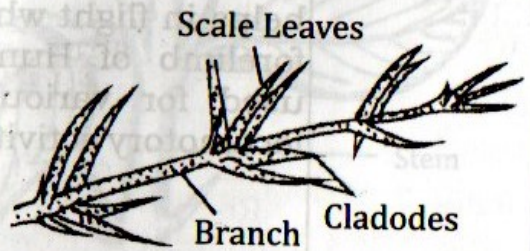
Organs of modification	Organs of adaptations	Examples
Scale leaves of onion and spines of opuntia (Prickly pear)	• The scale leaves of Onion and spines of opuntia are homologous to one another as both are modified leaves. • Scaly leaves of onion are thick, fleshy and stores water and food, while spines of opuntia are sharp and defense in funtion. • Organs of these plants show different functions but same origin hence called Homologous organs. • These organs exhibit divergent evolution.	 **Scale leaves of Onion**  **Spines leaves of Opuntia**

Organs of modification	Organs of adaptations	Examples
Tendrils of Passion flower and thorns of Bougainvillea	The tendril of passion flower and thorns of bougainvillea originate from axillary bud, but they are different in structure and function. Tendrils are used for holding and thorns for defensive.	 Tendrils of passion flower  Thorns of Bougainvillea
Tendrils of *Vitis* and thorns of *Carissa*	Tendrils of *Vitis* and thorns of *Carissa* originate from terminal bud, but they are functionally different	 Tendrils of Vitis  Thorns of Carissa

Homologous Organs in Animals

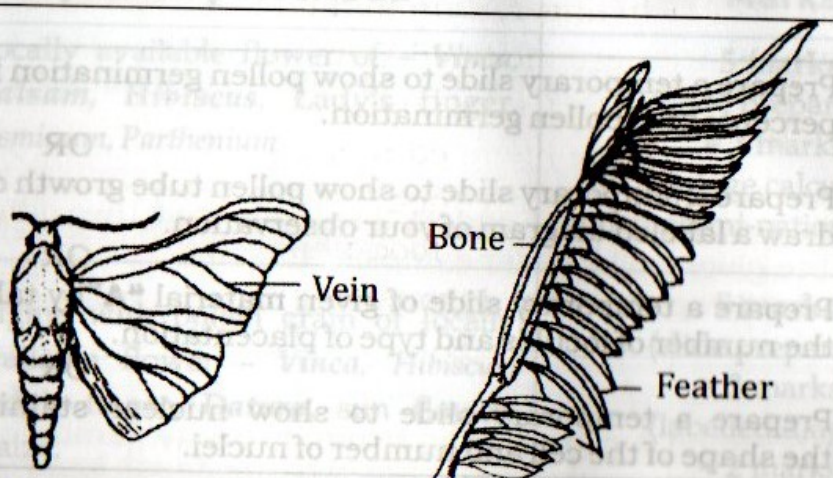
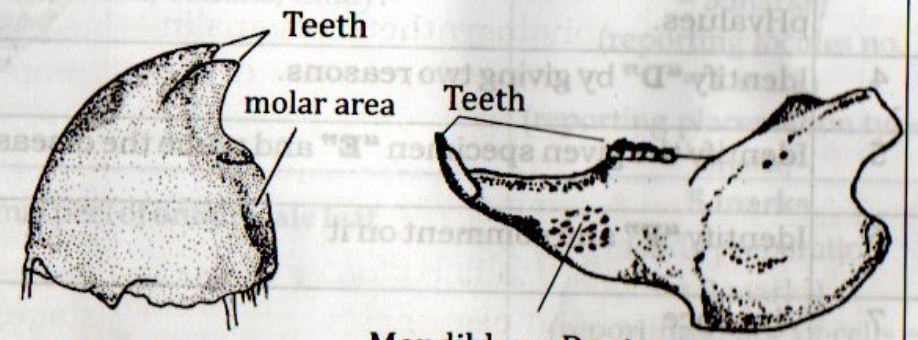
Organs of modification	Organs of adaptations	Examples
Fore-limbs of vertebrates (Bat & Human)	The forelimbs of a Bat and a Human are homologous to one another as they have similar anatomical structures but perform different functions. b. The bones namely humerus, radius, ulna, carpals, metacarpals and phalanges are found in both Bat and Human. c. The forelimb of Bat helps in flight while the forelimb of Human is used for various non locomotory activities.	 **FORE LIMB SKELETON OF (A) BAT (B) HUMAN**

EXERCISE - 24 : ANALOGOUS ORGANS IN PLANTS & ANIMALS

Organs of modification	Organs of adaptations	Examples
Modified underground stem and modified roots.	• Ginger, Yam and Potato are modified stems, but the Carrot and Radish are the modified root. • Both are modified for storage of food hence they are analogous organs.	 Rhizome of Ginger Modified root of Carrot
Phylloclade, Cladode and Leaves	• Phylloclade, Cladode and Leaves are the photosynthetic organs in different plants, in *Opuntia* stem is modified for photosynthesis and in *Asparagus*, the inter-node is modified for photosynthesis.	 Phylloclade of Opuntia  Cladode of Asparagus

Analogous organs In animals

Organs of similar function and appearance, but different in structure and origin are called analogous organs.

Organs of modification	Organs of adaptations	Examples
The wings of an insects and of a bird :	The wings of an insects and of a birds are analogous organs the perform same function but anatomically different structures The wings of insects is lifeless structure. it is supported by veins covered with chitinous exokeleton. The wing of bird is supported by fore limb and is covered with feathers they help in flight.	 **Wing of insect** **Wing of a Bird**
Mandible of Cockroach and lower jaw of Rabbit	a . The mandible of a Cockroach and the lower jaw of Rabbit are analogous to one another as they perform a similar function but are structurally different. b. The mandible of Cockroach is chitinized (sclerotized) strucuture while the lower jaw of Rabbit is bony. c. Both help in cutting and grinding the food.	 **Mandible of Cockroach** **Lower jaw of rabbit**

II YEAR PUC PRACTICAL EXAMINATION

TIME : 2 HOURS **SUBJECT : BIOLOGY (36)** **MAXIMUM MARKS : 30**

Sl No.	QUESTIONS	MARKS ALLOTTED
1	Prepare a temporary slide to show pollen germination from the given material **"A"** and calculate the percentage of pollen germination. OR Prepare a temporary slide to show pollen tube growth on the stigma from the given material **"A"** and draw a labeled diagram of your observation. OR Prepare a temporary slide of given material **"A"** by taking a transverse section of ovary and report the number of locules and type of placentation. OR Prepare a temporary slide to show nuclear staining from the given material **"A"** and report the shape of the cell and number of nuclei.	***5 marks***
2	Prepare a temporary slide of the given material **"B"** and identify any one stage of mitosis and draw a labeled diagram of the stage observed.	***5 marks***
3	Conduct a suitable test for the given soil/water samples **"C1"** and **"C2"** and report the pHvalues.	***3 marks***
4	Identify **"D"** by giving two reasons.	***3 marks***
5	Identify the given specimen **"E"** and name the disease it causes	***2 marks***
6	Identify **"F"** and comment on it	***2 marks***
7	Viva voce	***4 marks***
8	Practical Records (Both Botany and Zoology)	***6 marks***

SCHEME OF EVALUATION

TIME : 2 HOURS BIOLOGY **MAXIMUM MARKS : 30**

Q No.	Question Type	Syllabus prescribed	Marks
1. Micropreparation / Slide preparation (Any one expt. Selection by lottery system)	A_1) Prepare a temporary slide to show **invitro pollen germination** from the given material "**A**" and **calculate** the percentage of pollen germination.	• Locally available flower of – ***Vinca, Balsam, Hibiscus***, Lady's finger, *Jasminum, Parthenium*	**5 marks** (slide preparation = 3 marks) (percentage calcula-tion of pollen germi-nation = 2 mark)
	A_2) Prepare a temporary slide to show **pollen tube growth on the stigma** from the given material "**A**" and draw a labelled **diagram** of your observation.	• Stigma with pollen grain of locally available flower – ***Vinca***, *Hibiscus*, Lady's finger, ***Datura***, sun flower, maize.	**5 marks** (slide preparation = 3 marks) (labelled diagram = 2 marks)
	A_3) Prepare a temporary slide of given material "**A**" by taking a **transverse section** of ovary and **report** the number of locules and type of placentation.	Locally available flower of - pea, ***Hibiscus***, Lady's finger, ***Datura***, Mustard, Sunflower, *Tridax*, Tomato, Chilly.	**5 marks** (slide preparation = 3 marks) (reporting locules no. = 1 mark) (reporting placentation type = 1 mark)
	A_4) Prepare a temporary slide to show **nuclear staining** from the given material "**A**" and **report** the shape of the cells and the number of nuclei.	Epidermal peel of **onion scale leaf**	**5 marks** (slide preparation = 3 marks) (reporting shape of cells = 1 mark) (reporting number of nuclei = 1 mark)

2	Cytology: Prepare a temporary slide of given material "**B**" and **Identify** any one stage of mitosis and draw a labelled **diagram** of the stage observed.	Fresh / preserved onion root tip (Any one stage of mitosis – interphase / prophase / metaphase / anaphase / telophase.	**5 marks** (slide preparation = 3 marks) (identifying any one stage = 1 mark) (labelled diagram = 1 mark)
3	Ecology experiment: Conduct a suitable **test** for the given soil / water samples "**C1**" and "**C2**" and **report** the p^H values.	Two different samples of soil solution. (OR) Two different water samples	**3 marks** (reporting pH of sample A = 1½ mark) (reporting pH of sample B = 1½ mark)
4	Identification & commenting (from histology / embryology / cytology): **Identify** the given slide / model / photograph **"D"** giving atleast **two** suitable reasons.	Embryological or histological or cytology slide / photograph / model of any one of the following- V.S of ovule showing female gametophyte, T.S of mammalian testis, T.S of mammalian ovary, V.S of blastocyst (blastula), stages of meiosis (prophase-I & II, metaphase-I & II, anaphase-I & II, telophase-I & II).	**3 marks** (identification = 1 mark) (any two comments = 2 mark)
5	Identification & commenting (from human diseases): **Identify** the given slide / specimen / model / photograph **"E"** and **name** the disease it causes.	Slide / specimen / photograph / model of any one of the following disease causing – *Entamoeba*, *Plasmodium* (signet ring stage), *Ascaris* male/ female, *Trichophyton*.	**2 marks** (identification with scientific name = 1 mark) (naming the disease = 1 mark)

6	Identification & commenting (from ecology / evolution / genetics): **Identify** the given slide / specimen / photograph **"F"** and comment on it.	Slide / specimen / photograph / model of any one of the following: i) Plants showing **xeric adaptations:** T.S of *Nerium* leaf (sunken stomata, thick cuticle, multilayered epidermis), *Opuntia* (stem-phylloclade, leaves-spines), *Acacia* (petiole-phyllode), *Asparagus* (stem branch-cladode) *Bryophyllum* (leaf-reproduction). ii) Animals showing xeric adaptations: Camel, desert rat, squirrel. iii) Plants showing hydric adaptations: *Hydrilla, Eichornia, Nelumbo, Vallisnaria, Utricularia* iv) Animals showing hydric adaptations: Fish, prawn, frog. v) **Homologous Organs**: a) Tendrils of passion flower + thorns of pomegranate b) Tendrils of *Vitis* + thorns of Carissa c) Scale leaves of onion + spines of cactus d) Fore limb bones of frog or lizard or bat + fore limb bones of human vi) **Analogous Organs**: a) Stem tendrils of pea + leaf tendrils of *Vitis.* b) Thorns of pomegranate + spines of opuntia (prickly pear). c) Rhizome in ginger or tuber in potato + conical root in carrot or fusiform root in radish or napiform root in turnip or fasciculated roots in *Dhalia* d) Phylloclade in *Opuntia* or cladode in *Asparagus* + leaf of *Hibiscus* or mango. e) Wing of dragonfly / cockroach / butterfly + wing of bird. f) Mandible of cockroach + lower jaw of rabbit. Any one **Pedigree chart**.	**2 marks** (identification = 1 mark) (any two comments = 1 mark)
7	**VIVA VOCE (4 Qns)**	(2 Qs by external + 2 Qs by internal to be asked)	**4 marks**
8	**PRACTICAL RECORD**	(Botany + Zoology) 91% expts.	**6 marks**

	% of experiments performed and recorded	**Maximum marks to be awarded**
1	91%	6 - more than 23 practical recorded
2	81% to 90%	5 - more than 21 practical recorded
3	71% to 80%	4 - more than 18 practical recorded
4	Between 41% and 70%	3 – more than 11 practical recorded
5	40% & below 40%	0

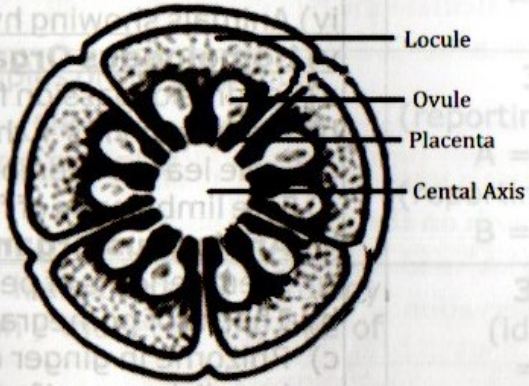
Model Answer Paper

Subject: **BIOLOGY** Student No: | | | | | | | | | | | |

1. Micropreparation:

A_3) Prepare a temporary slide of given material "A" by taking a transverse section of ovary and report the number of locules and type of placentation. (5 marks)

Source of ovary : Hibiscus
Number of Locules: Five, so called pentalocular
Type of placentation: Axile placentation
(OR)
Source of ovary : Datura
Number of Locules: Four, so called tetralocular
Type of placentation: Axile placentation
(OR)
Source of ovary : Cucurbita
Number of Locules: Three, so called trilocular
Type of placentation: Parietal placentation
(slide preparation = 3 marks)
(reporting locules no. = 1 mark) (reporting placentation type = 1 mark)



Axile Placentation

T.S. OF HIBISCUS OVARY

2. Cytology: **Prepare a temporary slide of given material "B" and Identify any one stage of mitosis and draw a labelled diagram of the stage observed.** **(5 marks)**

Identification: The stage identified by me in the slide preparation of material "B" is anaphase of mitosis.

(slide preparation = 3 marks) (identifying any one stage = 1 mark)
(labelled diagram = 1 mark)

Anaphase

3. **Ecology experiment: Conduct a suitable test for the given soil / water samples "C1" and "C2" and report the p^H values.** **(3 marks)**

Measurement of pH of the given samples C_1 and C_2

Sample	pH value determined by Broad range indicator paper	pH value determined by Narrow range indicator paper	pH value determined by Universal Indicator solution	Sample type
C1	**6.0**	**5.5**	**5.5**	**Acidic**
C2	**8.0**	**7.0**	**7.0**	**Neutral**

Result: Sample C1 is identified as Acidic solution
Sample C2 is identified as Neutral solution
(reporting pH of sample A = 1½ mark) (reporting pH of sample B = 1½ mark)

Q4. Identification & commenting (from histology/embryology/cytology): Identify the given slide/model/photograph "D" giving atleast two suitable reasons. **....3 marks**

Identification: The given slide "D" is identified by me as T. S of mammalian ovary showing graffian follicle.

Reasons:

1. Ovary is lined with germinal epithelium (primordial germ cells).
2. Section shows many immature follicles (primary, secondary and tertiary) and few mature / graffian follicles.
3. Graffian follicle shows a cavity - antrum with mature ovum (secondary oocyte stage)
4. A post ovulatory phase of follicle called corpus luteum is seen.
5. Degenerating stage of follicle called corpus albicans is lightly seen.

(identification = 1 mark) (any two comments = 2 mark)

5. **Identification & commenting (from human diseases): Identify the given slide/specimen/model/photograph "E" and name the disease it causes.** **....2 marks**

Identification: The given beautiful photograph "E" is identified by me as intracellular protozoan parasite called Plasmodium vivax

in Signet ring stage in human RBC.
Disease caused: Plasmodium causes Malaria.
(identification with scientific name = 1 mark) (naming the disease = 1 mark)

6. **Identification & commenting (from ecology / evolution / genetics): Identify the given slide / specimen / model / photograph "F" and comment on it.** **....2 marks**

Identification: The given beautiful photograph "F" is identified by me as Nelumbo nucifera (Lotus), showing hydric adaptations.

Comments / Reasons.

1. Petiole is long, slender and spongy & are flexible and withstand water currents.
2. Petiole and other parts are made up of aerenchyma tissue with air pockets for buoyancy.
3. Leaf blade is broad with stomata only on upper epidermis (epistomatic).
4. Waxy cuticle on the surfaces prevent wetting and clogging of stomata by water.
5. Stomata present only on upper surface of leaf, so called epistomatic.

It is a fresh water floating leaved, anchored hydrophyte. The stem is rhizomatous and grows horizontally under the mud.
(identification = 1 mark) (any two comments = 1 mark)

7. **Viva Voce: (2 questions by external 2 by internal to be asked–one mark each) .....4 marks**

Eg: 1) Mention any one function of tapetum.
Provide nourishment to developing pollen grain; secrete sporopollenin to form exine around pollen grain

2) What is Sporopollenin ?
The organic resistance substance present in exine of pollen grain

3) In the root tip squash that you have prepared, some of the cells are not seen with nucleus. Why ?
Because cells are found with metaphase and anaphase stage of cell cycle.

4) What is the is the range of pH value of a basic substance ?
pH = 7.1 to 14 (more than 7 upto 14)

8. **Practical Record: (Botany+Zoology) duly certified by lab in charge & principal. (6 marks)**

Note:
Question type with material 'A' for Experiment – 1 is to be selected by lottery.
Student must need to bring Practical record (duly certified & bound) at the time of exam.
Student must need to bring materials like slide, cover slips, fresh razor blade, '0' number painting brush, soft cotton cloth piece, needle, forceps, syrup bottle cap (for pH sample), flowers etc., at the time of practical examination with out which he / she can not be allowed to do the practical exam.
The above said scheme and syllabus copy is not allowed to the practical examination.

POSSIBLE VIVA VOCE QUESTIONS

REPRODUCTIVE PARTS OF A FLOWER

1. **What is flower?**
   Flower is the site of sexual reproduction.
2. **What is placentation ?**
   The mode of arrangement of ovules within the ovary.
3. **Mention any two types of placentation.**
   Axile, marginal, parietal, basal and free central
4. **What is common name of Hibiscus.**
   China rose (dhasavala / mandara in Kann)
5. **What are locules ?**
   The chamber/s present inside the ovary containing ovules
6. **Name the type of placentation found in Hibiscus / Datura.**
   Axile placentation.
7. **How many locules are present in ovary of Hibiscus / Datura**
   Hibiscus - pentalocular
   Datura – bilocular / tetralocular
8. **How many ovules are present in each locule of Hisbiscus ovary**
   Tow ovules
9. **What is Marginal placentation**
   Here the ovules are attached to the margin of the ovary wall Eg., pea
10. **What is Axile placentation?**
    Here the ovules are attached to the central axis. Eg., Hibiscus
11. **What is Parietal placentation?**
    Here the ovules are attached along the peripheral region of the ovary wall. Eg., papaya, cucumber
12. **What is Basal placentation**
    Here the ovules are attached to the base of the ovary wall. eg., sunflower.
13. **Which is most common type of placentation in plants?**
    Axile
14. **What are present inside the locules?**
    Ovules.

POLLEN GERMINATION ON NUTRIENT MEDIUM.

1. **What is pollen grain ?**
   Microspore, which germinates into male gametophyte containing male gametes.
2. **Name the two walls of microspore.**
   Outer exine & inner intine
3. **What is pollen germination ?**
   Growth of pollen tube through germ pore.
4. **Which layer of sporangium secrets sporopollenin ?**
   Tapetum
5. **Mention any one function of tapetum.**
   Provide nourishment to developing pollen grain; secrete sporopollenin to form exine around pollen grain.
6. **During germination which part of the pollen grain emerges out as a tube ?**

Intine (inner wall of pollen grain)

7. **Where do the germ pores are present?**
   In exine (outer wall of pollen grain)
8. **From where does pollen tube emerge?**
   Through the germ pores
9. **Exine in pollen grain is made of what?**
   Sporopollenin
10. **Name the organic resistance substance present in exine of pollen grain.**
    Sporopollenin.
11. **Name two cells of mature pollen grain.**
    Vegetative cell and generative cell.
12. **Which cell emerges out as pollen tube?**
    Vegetative cell / tube cell
13. **How many pollen tubes emerge from a pollen grain?**
    In Vinca rosea three pollen tubes emerge (may range from 1 to 3)
14. **What does the pollen tube carry ?**
    Pollen tube carries one vegetative nucleus and two male gametes.
15. **Which medium you have used for pollen germination ?**
    10% sucrose solution
16. **How do you prepare 10% sucrose solution ?**
    By dissolving 10g sucrose (table sugar) in 100 ml water.
17. **Why do we use sucrose solution as a medium for pollen germination ?**
    To provide a similar condition as that of compatible stigma, where a sweet substance acts as a nourishing medium which induces pollen germination.
18. **What is the common / local name of Vinca rosea ?**
    Kasi kanagale / periwinkle
19. **Where does the pollen grain germinate naturally?**
    On compatible stigmatic surface.
20. **What stimulates the growth of pollen germination?**
    Sugary substance present on the stigma stimulates the pollen germination.
21. **Which layer of pollen wall grows out as pollen tube?**
    Intine

POLLEN TUBE GROWTH ON STIGMA

1. **What stain is used to study pollen tube growth in stigma?**
   Cotton blue / safframin
2. **What is the purpose of boiling stigmas before staining ?**
   To soften the tissues
3. **Name the stain used in your preparation to show pollen tube on stigma.**
   Cotton blue stain / saffranin stain / methyline blue stain
4. **What is porogamy ?**
   Entry of pollen tube into the ovule through micropyle.

STAINING OF NUCLEIC ACID

1. **What is the position of nucleus in the cells of onion peel ? Why ?**
   Peripheral, due to the presence of central large vacuole.
2. **Why some cells seems to be without nucleus and some with two nuclei ?**
   Due to the rupturing / damage of cells.

3. **What is a stain / dye ?**
   A substance used to impart color to tissues or cells / cellular particles in laboratory.
4. **Name any two stains used in micro preparation / nuclear staining.**
   Aceotocarmine, Acetoorcein, safranin, eosin.
5. **What is an acidic stain ?**
   Acid stains bind to positively charged molecules and impart colour to it.
6. **Give an example for acidic stain.**
   Eosin
7. **What is a basic stain ?**
   Basic stains bind to negatively charged molecules and impart colour to it.
8. **Give an example for basic stain.**
   Aceotocarmine, Acetoorcein, safranin, (others like crystal violet, methylene Blue, Hematoxylin and basic fuchsin).
9. **Why do we use acetocarmine stain in micro preparations ?**
   Acetocarmine is a basic dye which combines with nucleic acid and make the nucleus appear red.
10. **What is the botanical name of onion ?**
    Allum cepa
11. **Mention the diploid number of chromosomes in onion.**
    2n = 16, n = 8

MITOSIS IN ONION ROOT TIP

1. **Why 1 N Hcl is used in mitotic squash preparation?**
   To dissolve the middle lamella present between the cells
2. **What is the significance of mitosis?**
   Growth and repair of body.
3. **Which stain is used to study mitosis?**
   Aceto caramine or Aceto orcein stain.
4. **Why is root tip used to show mitosis division?**
   Root tip has meristamatic cells.
5. **What would be the shape of the chromosomes during Anaphase stage?**
   Metacentric ..... V shaped
   Sub-meta centric ..... J shaoed
   Telo centric ..... I shaped
6. **How do you identify Metaphase stage?**
   Chromosomes are aligned on metaphase plate / Equatorial plate
7. **How do you identify Anaphase stage?**
   The centromeres face the poles and arms of the daughter chromatids face the equator.
8. **When does cytokinesis occur?**
   After telophase
9. **What is the function of spindle in dividing cells?**
   Movement of chromosomes.
10. **What is Cytokinesis?**
    Division of Cytoplasm.
11. **In which stage of mitosis chromosomes are at equatorial plane ?** - Metaphase
12. **In which stage of mitosis chromosome move towards opposite poles ?** - Anaphase
13. **What is the function of spindle in dividing cells?**
    movement of chromosomes.

STAGES OF MEIOSIS

1. **In which type of cell division chromosome number is reduced ?**
   Meiosis
2. **Why meiosis–I is called reductional division ?**
   Because the daughter cells receive half the number of chromosomes from mother cell (from diploid to haploid).
3. **What are meiocytes ?**
   The cells which undergo meiotic cell division
4. **In which cell (meiocytes) meiosis occurs?**
   Diploid germ cells like spermatogonial / oogonial cells / spore mother cells
5. **Mention the stages in prophase – I of meiosis.**
   Leptotene (leptonema), zygotene (zygonema), pachytene (pachynema), diplotene (diplonema) and diakinesis.
6. **What are bivalents ?**
   The paired homologous chromosomes during meiosis (zygotene).
7. **What are homologous chromosomes?**
   A pair of morphologically and genetically similar chromosomes derived from both male and female parents.
8. **What is synapsis ?**
   Pairing of homologous chromosomes lengthwise during zygotene.
9. **Define crossing over.**
   Exchange of genetic material between two non-sister chromatids of same homologous pair.
10. **In which stage of meiosis crossing over occurs?**
    Pachytene / Pachynema
11. **What is chiasma formation ?**
    X – shaped configuration seen at the region of crossing over in a bivalent.
12. **In which stage of meiosis chiasmata formation is seen ?**
    In Diplotene stage
13. **In anaphase – I, each chromosome shows how many chromatids ?**
    Two chromatids
14. **What is the significance of meiosis ?**
    Production of haploid gametes to ensure diploidy in next generation with recombination of characters.
15. **How many daughter cells are produced at the end of meiosis?**
    Four daughter cells are formed.

ANALYSIS OF PH OF SOIL & WATER SAMPLE

1. **What is pH ?**
   Negative logarithm of H^+ ion concentration.
2. **On what basis pH is determined ?**
   Based on more H^+ ion concentration (as acidic) or more OH^- ion concentration (as basic) in the substance.
3. **Name the Three methods of determination of pH value of water / soil sample.**
   1) indicator dye method
   2) electrometric method using ph meter &
   3) colorimetric method
4. **Name the method to test pH of water in your laboratory ?**
   Indicator dye method.

5. **Which is the most accurate method of pH determination ?**
electrometric method using ph meter
6. **What is the range of pH value of an acidic substance?**
pH = 1 to 6.9 (less than 7)
7. **Name the two types of pH indicator paper**
Narrow range (specific range) and broad range (wide range)
8. **What is the is the range of pH value of a basic substance ?**
pH = 7.1 to 14 (more than 7 upto 14)
9. **What is the highest value of pH ?**
pH 14
10. **What is the pH value of a neutral substance solution ?**
pH = 7
11. **What is the pH value of distilled water?**
Neutral pH (7)
12. **What is the pH of rain water ?**
Normal rain is slightly acidic, with pH 6-5.7 as it dissolves atmospheric CO_2

TEXTURE OF SOIL, TURBIDITY OF WATER, PLANT POPULATION DENSITY BY QUADRAT METHOD, & FREQUENCY BY QUADRAT METHOD

1. **Water logged soils are acidic. Why ?**
Due to bacterial activity, there is a formation sulfuric acids and oxides of iron compounds (pyrites)
2. **Why are the soil around mineral mining areas acidic?**
Oxides of mineral elements dissolve in water and form sulfuric and nitric acids
3. **What is the pH scale given in colour chart with broad (wide) range pH indicator paper given to you?**
pH 2.0 to pH 10.5
5 **What is the pH scale given in colour chart with narrow (specific) range pH indicator paper given to you?**
pH 5.0 to pH 7.5
6. **Name different types of soil**
(a) Red Latosol - Red in colour due to iron oxide
(b) Black Soil - Rich in iron and magnesium
(c) Alluvial Soil - It is mixture of clay, snad and silt.
(d) Desert Soil - Sand, porous and grey brown soil.
(e) Mountain soil - Very fertile, rich in humus.
(f) Laterite soil - Clay red coloured soil.
7. **Which soil is the best for crops?**
Loamy.
8. **What is Pedology?**
Study of soil.
9. **How the soil is formed?**
Soil is formed by decay and decomposition of complex organic compounds of dead plants, animals and other organisms.
10. **What does soil texture mean?**
It means relative proportions of different particles of soil.
11. **What is quadrat?**
Quadrat is a sample plot of specific size used for study of a population
12. **Why quadrat is usually selected of small size?**

In small units, individuals of different species can be counted easily and accurately.

13. **Why population of a species never remains constant?**
Population of a species may increase of decrease due to birth, death, migration etc.

FEMALE GAMETOPHYTE DEVELOPMENT

1. **What is the other name of ovule ?**
Megasporangioum
2. **Which part of the ovule is called female gametophyte ?**
Embryosac with nucellus
3. **How does the female gametophyte (embryosac) is formed ?**
Embryosac is formed from a megaspore mother cell through reduction division.
5. **Name the stalk of the ovule.**
Funicle
6. **Which part is known as the junction between ovule and funicle ?**
Hilum
7. **What are integuments ?**
Protective envelopes of ovule
8. **Name the cells present in mature embryosac.**
Egg with two synargids together called egg apparatus at micropylar end; two antipodals at chalazal end; two polar nuclei (polar cell) in the centre.
9. **Name the cell components of egg apparatus.**
One egg cell and two synergids.
10. **How many cells / nuclei does the mature female gametophyte possess ?**
7 cells with 8 haploid nuclei
11. **To which end of ovule synergids present ?**
Towards micropylar end
12. **Where do you find filiform apparatus ? Mention its significances.**
These are special cellular thickenings in synergids which help in guiding the pollen tube to enter into synergid and then towards egg cell.
13. **Name the cells present towards chalazal end of ovule.**
Antipodals
14. **What is the female gametophyte also called?**
Embryo sac
15. **Where does ovule develop?**
Inside ovary
16. **Each ovule has how many female gametophyte generally?**
One
17. **What is Monosporic type of development of female gametophyte?**
Method of development of embryosae or female gametophyte form a single megaspore, is called monosporic
18. **A typical angiosperm embryo sac at maturity, is _____ nucleated & _____ celled.**
8 , 7
19. **What is the function of polar nuclei ?**
Polar nuclei, after double furtilization develops in the endosperm.

GAMETOGENESIS

1. **Define gametogenesis**
   Formation of gametes from diploid germ cells in gonads
2. **Name the three events in the process of gametogenesis.**
   Phase of multiplication, phase of growth and phase of maturation.
3. **Define spermatogenesis.**
   The process of formation of haploid male gametes / spermatids from dipoid spermatogonial cells in testis.
4. **Name the connective tissue covering of testis.**
   Tunica albugenia
5. **What is spermatid ?**
   Non-functional, non-motile, inactive haploid male gamete.
6. **How many spermatids does a primary spermatocyte produce?**
   Four
7. **What are spermatozoa ?**
   Functional, motile, active haploid male gametes.
8. **Name the structural & functional units of testis /name the site of sperm production.**
   Seminiferous tubules
9. **Name the two types of cells found in / produced from germinal epithelium of seminiferous tubules.**
   Spermatogonia and sertoli cells
10. **What is the functional role of sertoli cells ?**
    Nourishing the sperms
11. **Name the hormone that stimulates spermatogenesis.**
    Androgens and ICSH (= LH)
12. **Name the cells that secrete androgens (testosterone).**
    Leydig cells / Interstitial cells
13. **At puberty how many primary follicles are left in each ovary.**
    Only 60,000 to 80,000 primary follicles
14. **When does meiosis–II completes during the process of oogenesis ?**
    After the entry of sperm into secondary oocyte.
15. **What is graffian follicle ?**
    The mature follicle with well developed antrum and secondary oocyte / mature ovum.
16. **What is antrum ?**
    The fluid filled cavity present in graffian follicle.
17. **Name the hormone secreted by graffian follicle.**
    Oestrogens
18. **What is ovulation ?**
    Releasing of secondary oocyte / oocyte / ovum into the fallopian tube by the rupturing of graffian follicle.
19. **At which stage of follicular development is ovum released ?**
    Graffian follicle stage
20. **What is corpus luteum ?**
    Post-ovulatory follicle, a temporary endocrine gland producing progesterone.
21. **Name the hormone which stimulates ovulation.**
    Lutenising hormone (LH)
22. **Name the hormone produced by corpus luteum.**
    Progesterone
23. **What is corpus albicans ?**
    The degenerating stage of corpus luteum when the egg

is not fertilized.

24. **Name the hormones secreted by corpus luteum & Leydig cells?**
Progesteron - Testosteron (Androgen)
25. **Name male sex organ or gonad of mammals.**
Testis
26. **Name the female sex organ or gonad of mammals.**
Ovary
27. **Name the nutritive cells present in the testis.**
Sertoli cells or Sustentacular cells
28. **Name the male sex hormone secreted by Leydig cells.**
Androgens

BLASTULA OF MAMMALS

1. **What are the main structures that you observe in T.S. of blastula ?**
Blastoderm (trophoblast layer), inner cell mass, blastocoel cavity, zona pellucida, cells of Rauber.
2. **What is Blastocyst ?**
Single germ layered embryo of mammals / human.
3. **Name the outer single cell layer (germ layer) in blastocyst**
Trophoblast / Blastoderm
4. **Name the cells present in the outer cell layer of blastocyst.**
Trophoblast cells
5. **Name the fluid filled cavity present in blastocyst.**
Blastocoel
6. **Name the earlier stage of embryo with solid ball of cells prior to blastula.**
Morula
7. **Name the later stage of embryo next to blastula.**
Gastrula
8. **Name the part of the blastula which develops into embryo proper.**
Inner cell mass
9. **What are cells of Rauber ?**
These are trophoblast cells that cover / line the inner cell mass
10. **What is implantation ?**
Attachment / anchorage of blastocyst to the wall of uterus.

COMMON DISEASE CAUSING ORGANISMS

1. **To which phylum Entamoeba belongs ?**
Phylum Protozoa
2. **Name the locomotory organelle of Entamoeba.**
Single pseudopodium / Lobopodium
3. **Name the disease caused by Kntamoeba.**
Amoebic dysentery
4. **Which is the vector for malarian parasite ?**
Female Anopheles mosquito
5. **Why Plasmodium is called digenetic parasite ?**
Because it requires two hosts to complete its life cycle.
6. **Name the two hosts in which plasmodium completes its life cycle.**
Man (primary host) and mosquito (secondary host).
7. **Name the phylum to which Ascaris belongs.**
Phylum Aschelmintes
8. **What is the scientific name of round worm ?**

Ascaris lumbricoides

9. **Why Ascaris is called monogenetic parasite ?**
Because it requires only one host to complete its life cycle.

10. **Name the host organism for round worm.**
Human, more common in children.

11. **Name the site in which round worm is inhabited in man as a parasite.**
In small intestine

12. **How can you identify male Ascaris ?**
By characters like short body, curved posterior end with penial setae (copulatory spicules)

13. **How can you identify female Ascaris ?**
By characters like long body, straight posterior end without penial setae,

14. **Name the Malarial parasite.**
Plasmodium

15. **What are the symptoms of Malaria?**
High fever and chills, shivering following by lot of sweating, Loss of appetite, Sleeplessness

16. **Name the vector (carrier) for plasmodium?**
Female Anopheles mosquito

17. **Which is the diagnostic stage of the plasmodium?**
Signet ring stage

18. **Which is the infective stage of plasmodium?**
Sporozoite

19. **Which cells of malarial patient is infected by plasmodium?**
RBC

20. **Name the disease caused by ascaris?** - Ascariasis

21. **Symptoms of Ascariasis?**
Irregular bowel, Indigestion, Loss of appetite, Abdominal pain, Anaemia, Fatigue, Vomitting.

22. **What is the causative agent of Ring worm disease?**
Trichophyton

23. **What does Ring worm feed on?**
Keratin of skin and nails

24. **Symptoms of Ring worm disease.**
Ring shaped fluid filled lesions in skin, scalp, Itching Nails decolourise, Nails become brittle

25. **Symptoms of Amoebiasis**
The symptoms of this disease are frequent loose, mucus filled watery stools, abdominal pain and spasms.

26. **Name the two forms of parasite entamoeba.**
Trophozoit and cyst forms.

27. **Who discovered Malaria?**
Sir Ronald ross

28. **Name the causative organism of ringworm.**
Trichophyton or Microsporum or Epidermophyton

29. **Name the fungus that causes Athlete's foot or Barber's itch or ring worm.**
Trichophyton rubrum

CONTROLLED POLLINATION

1. **What is emasculation ?**
Removal of stamens before anthesis in a bisexual flower to obtain female parent.

2. **What is the significance of emasculation ? (OR) Why is emasculation is performed before anthesis ?**
To obtain female parent and to prevent self pollination and to promote cross pollination to obtain hybrid

progeny.

3. **What is bagging? What is its significance ?**
Covering the emasculated flower with a polythene bag to protect the stigma from contamination with undesirable pollen.
4. **What is cross breeding / artificial hybridization? Mention its significance.**
Crossing two genetically different parents of animals / plants of same species and often genera to get hybrids of desired qualities.
5. **How can you select female parent from the plants bearing bisexual flowers ?**
By emasculation using a pair of forceps.
6. **Name the material the bag made of used in bagging technique.**
Butter paper / polythene bag
7. **When do bagging is performed during hybridization?**
Before the flowers open and after cross pollination with desired pollen.
8. **What are the advantages of using a bag containing minute pores ?**
To ensure proper exchange of gases and transpiration.

MENDEL'S LAW OF SEGREGATION

1. **Name the plant selected by Mendel to carry out hybridization experiments.**
Pisum sativum / Garden pea plant
2. **What are alleles ?**
Alternate forms of genes at same gene loci in homologous pair controlling the same character.
3. **What is phenotype ?**
Externally expressed character of a gene.
4. **What is genotype ?**
Genetic constitution of a said character in an individual.
5. **Name the organisms which are similar phenotypically but different genotypically.**
Heterozygous.
6. **Define heterozygous condition.**
The organism with non-identical alleles for a said character. Eg : heterozygous tall = Tt
7. **What is a hybrid ?**
Genetically different individual obtained by crossing of two parents with different traits.
8. **Define monohybrid cross.**
A cross between two genetically different parents to study the inheritance of one pair of contrasting characters.
9. **Give an example for monohybrid cross.**
A cross b/w homozygous tall pea plant (TT) and homozygous dwarf pea plant (tt).
10. **What is a true breeding line ?**
A plant line when continuously self pollinated shows the stable trait inheritance through several generations.
Eg: pure tall (TT) or pure dwarf (tt)
11. **Write the phenotypic ratio of monohybrid cross.**
3 : 1
12. **Write genotypic ratio of monohybrid cross.**
1 :2 : 1
13. **What is the phenotypic & genotypic ratio when**

hybrid peas for tall are self pollinated ?
Phenotypic ratio = 3 : 1
Genotypic ratio = 1 :2 : 1

14. **Name the first law of Mendel.**
Law of segregation (OR) Law of purity of gametes.

15. **State the Mendel's 1st law of heredity.**
"In a hybrid the two alleles for a pair of contrasting characters do not blend / cross over and they segregate during gamete formation".

16. **Mendel's first law is called "Purity of gametes." Give reason.**
Because the gametes produced by F_1 hybrid receive either 'dominant' gene or 'recessive' gene, but not both due to segregation of alleles.

MENDEL'S LAW OF INDEPENDENT ASSORTMENT

1. **Name the units of inheritance.**
Genes / factors

2. **Define dihybrid cross?**
A cross b/w two genetically different parents to study the inheritance of two pairs of contrasting characters.

3. **Give an example for dihybrid cross.**
A cross b/w pea plant with round yellow seeds and pea plant with wrinkled green seeds.

4. **Mention the phenotypic ratio of a dihybrid cross.**
9 :3 : 3 : 1

5. **What is phenotypic ratio, when hybrid peas for yellow round seeds are self-pollinated?**
9 :3 : 3 : 1

6. **Name the second law of Mendel.**
Law of Independent assortment

7. **State the Mendel's 2nd law of heredity.**
"In a hybrid the pairs of alleles for two / more pairs of contrasting characters assort independent of one another during gamete formation". (OR)
"When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters."

8. **In a dihybrid cross, F_2 generation contains two new varieties. Give reason.**
Due to independent assortment of alleles / genes during gamete formation in a dihybrid.

9. **How many types of gametes are produced by dihybrid parent ?**
Four gametes (2^2)

PEDIGREE CHARTS

1. **What is pedigree analysis / pedigree chart ?**
Study and symbolic representation of inheritance of a particular trait through several generations of a family tree.

2. **What are the uses / significances of pedigree analysis ?**
- Useful in the study of pattern of inheritance of a specific trait.
- Useful in the study of a genetic abnormality / disease.
- Used to study mutations in a family history.

3. **What are siblings (sibs)?**
Offsprings of same parents. (OR) The progeny having same maternal & paternal parents.

4. **What are fraternal twins ?**
Dizygotic twins (non identical twins): Produced from two different eggs / zygotes.
5. **What are identical twins ?**
Identical twins (= Monozygotic): produced from same / single zygote.
6. **What does the arrow mark in pedigree chart indicates ?**
Arrow mark indicates subject , the person in the family which form the source of information (interviewed) about the trait.
7. **Give few examples for monogenic traits.**
Tongue rolling, widow's peak, blood groups, red-green colourblindness, dimple cheek, hypertrichosis of ear, Hitch-hiker's thumb etc.
8. **Name any two autosome linked dominant traits.**
Rolling of tongue, widow's peak (V-shaped hair line in forehead), dimple cheek, brachydactyly (shoprtening of fingers / toes), polydactuly (additional digits), clubbed thumb & fingers, etc.
9. **Name any two autosome linked recessive traits.**
Albinism, fused ear lobes, sickle cell anaemia, cystic fibrosis
10. **What is cystic fibrosis ?**
Disease of lungs, pancreas, liver and intestine, common in Europe.
11. **Name any two sex-linked (X-linked) dominant traits in humans.**
Duchene Muscular Dystrophy (oro-facial-digital syndrome – absence of teeth, cleft tongue, mental retardation), syndactyly (two or more digits are fused together),
12. **Name any two sex-linked (X-linked) recessive characters / diseases in humans.**
Red-green colour blindness, Haemophilia & Night blindness.
16. **Colourblindness (sex linked recessive character) is more prominent in males than in females.**
Because of Hemizygosity ie, the recessive gene for colourblindness is present only in X-chromosome and expresses itself in male even in the absence of its partner gene in 'Y'.
17. **Why sex linked inheritance is called criss - cross inheritance ?**
Because the genes located on X - chromosome of father passes to the grandson through carrier daughter.
18. **What is sickle cell anaemia?**
It is an autosomal genic disorder, where the RBCs are abnormal sickle shaped.
19. **Name the amino acid, which replaces the Glutamic acid in sickle cell anaemia.**
Valine
20. **What are Holandric genes?**
The sex - linked genes / characters associated with Y–chromosome.
21. **Give an example for Y-linkage in man.**
Hypertrichosis (hairy ears), Ichthyosis (fish like scaly skin)
22. **What is hypertrichosis?**
It is a Y-linked recessive character in man characterized by the development of hair from the ear pinna.

ECOLOGICAL ADAPTATIONS OF XERIC & HYDRIC ANIMALS

1. **What are xerophytes?**
   Plants living in dry or desert condition
2. **Name any four xerophytes.**
   Opuntia, Nerium, Acacia, Bryotphyllum, Zizyphus, Agave, Calotropis
3. **How do Xerophytes reduces transpiration?**
   Thick cuticle, reduced leaves, sunken stomata, multilayered epidermis, motor or buliform cells present
4. **What is the importance of succulents leaves & stem for xerophyte?**
   Storage of water
5. **Which part of opuntia plant is modified into spines why?**
   Leaves to reduce transpiration
6. **Name the stem modification in cactus.**
   Phylloclade
7. **What are hydrophytes?**
   Plants living in water.
8. **Name any four hydrophytes.**
   Hydrilla, Vallisneria, Eichhornia (water hyacinth), Nelumbo (lotus), Jussieua, Pistia, Utricularia
9. **What are the stress faced by hydrophytes?**
   Abundance of water, defiency of light and oxygen, tearing and decaying effect of water.
10. **Hydrophytic plants have arenchyma for what?**
    Buoyancy (helps in floating)
11. **List the structureal/ physiological adaptation in hydrophytes**
    Arenchyma, Swollen petiole, Epistomatic / no stomata, Balancing roots = buoyancy
12. **Why fishes have streamlined body?**
    Because it offers least resistance while swimming.
13. **What is the function of swim bladder in a fresh water (bony) fish?**
    Swim bladder stores air which help the fishes to float on water at required depth
14. **List any four hydric animals**
    Frog, fish, prawn, hydra
15. **Name the plant/animal that exhibit xeric adaptation**
    Opuntia, acacia, bryophyllum, nerium, etc Cammel, Squirrel & Kangaroo rat

HOMOLOGOUS & ANALOGOUS ORGANS OF PLANTS AND ANIMALS

1. **What are homlogous organs ?**
   The organs of different organisms which are similar in origin and structure but different in appearance and function due to change in habitats.
2. **Give an example for homologous organs in animals.**
   Fore limbs of human and bat
3. **What are analogous organs ?**
   The organs of different organisms which are dissimilar in origin and structure but similar in appearance and function due to similar habitat.
4. **Give an example for analogous organs in animals.**
   Wing of insect and wing of bird.

5. **What does homologous organs indicate ?**
Homologous organs indicate common ancestry and divergent evolution.
6. **What does analogous organs indicate ?**
Analogous organs indicate different ancestry and convergent evolution.
7. **What is adaptive radiation ?**
The process of evolution of different species from a common ancestor in a given geographical area through the development of adaptations is called Adaptive radiation
8. **Give an example for adaptive radiation.**
Variety of beaks of Darwin finches in Galapagos Islands due to different modes of feeding.
9. **What does forelimb of bat and wing of cockroach indicate in view of evolution ?**
Analogous organs
10. **What are homologous organs?**
Organs which have common origin and perform different function.
11. **What kind of evolution does homologous organs signify?**
Divergent evolution due to (different kinds of habitat)
12. **How are leaf tendril and leaf spines homologous?**
(Both arise at nodes, from axillary buds but perform different kinds of habitat)
13. **What is the difference in functions of leaf tendrils and spines?**
Spines - protect the plant from browsing animals
Tendrils - provide support for climbing.
14. **What are analogous organs?**
Organs which have similar function but different in basic plan and origin.
15. **Give an example for homologous organs in plants.**
Tendril and Thorn—Axillary bud modification
Tendrils and Thorns—Terminal bud modification
Tendrils and Bulbils—Floral bud modification
Scaly leaves and Spines—Leaf modification
16. **Name the stem modification in cactus.**
Phylloclade.
17. **What is cladode ?**
In Asparagus internodes are modified for photosynthesis are known as cladode.
18. **What is Bulbil ?**
Modified floral buds for reproduction in Agave.
19. **Give an example for analogous organs in plants.**
Stem tendrils and leaf tendrils Thorns and spines
Underground modification of root and stem
20. **Give an example for homologous organs in animals.**
Fore limbs of Vertebrates.
21. **Give an example for analogous organs in animals.**
Wings of insects and wings of birds
Mandible of cockroach and Rabbit.
22. **How did analogous organs originate?**
Due to convergent evolution. Due to similar habitat
24. **How are wings in different animals analogous organs?**
Because they have same function (flying) but their origin and basic structural plan is different.

Good luck